Underground Mine Development and Stopping Methods

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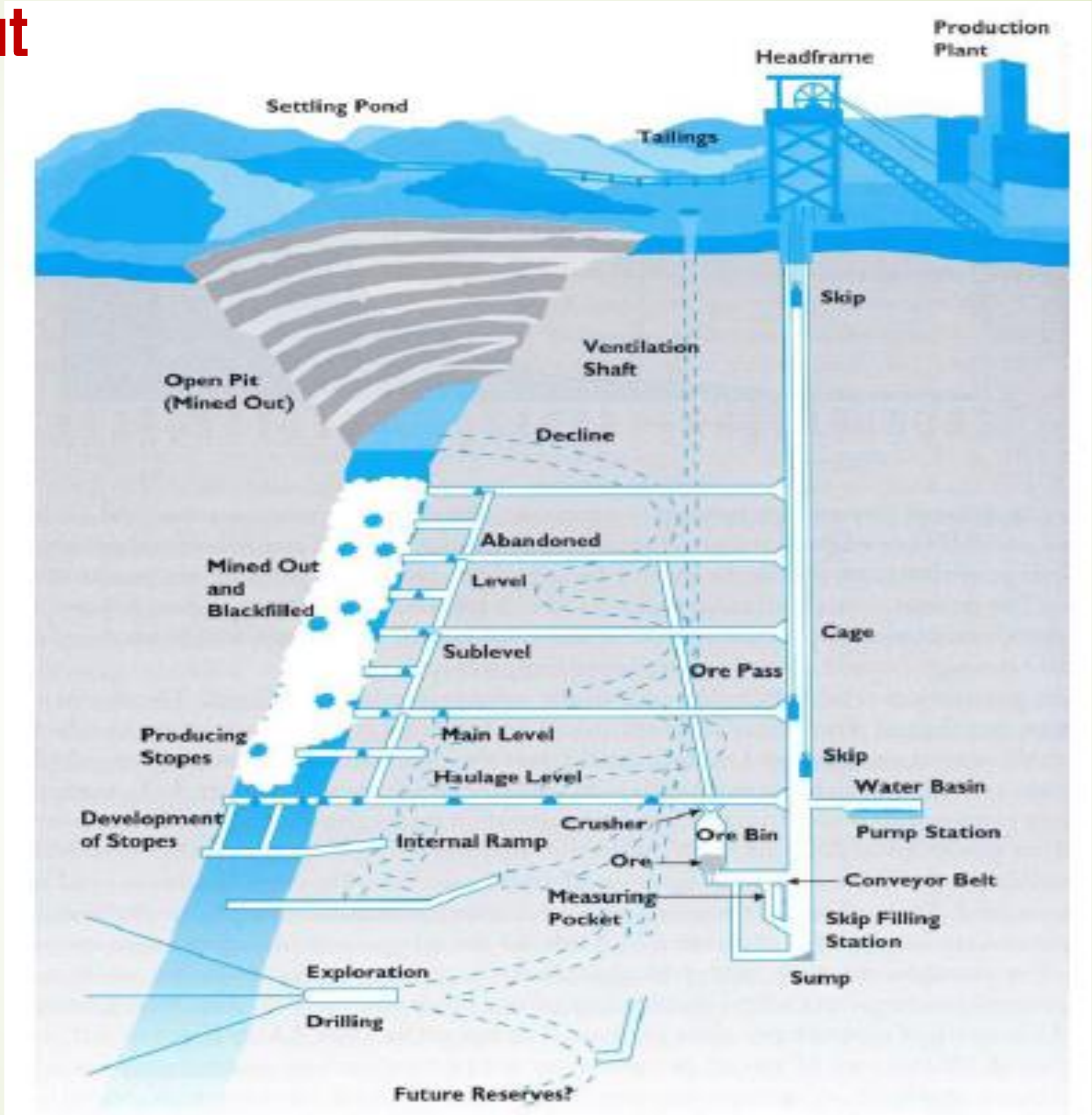
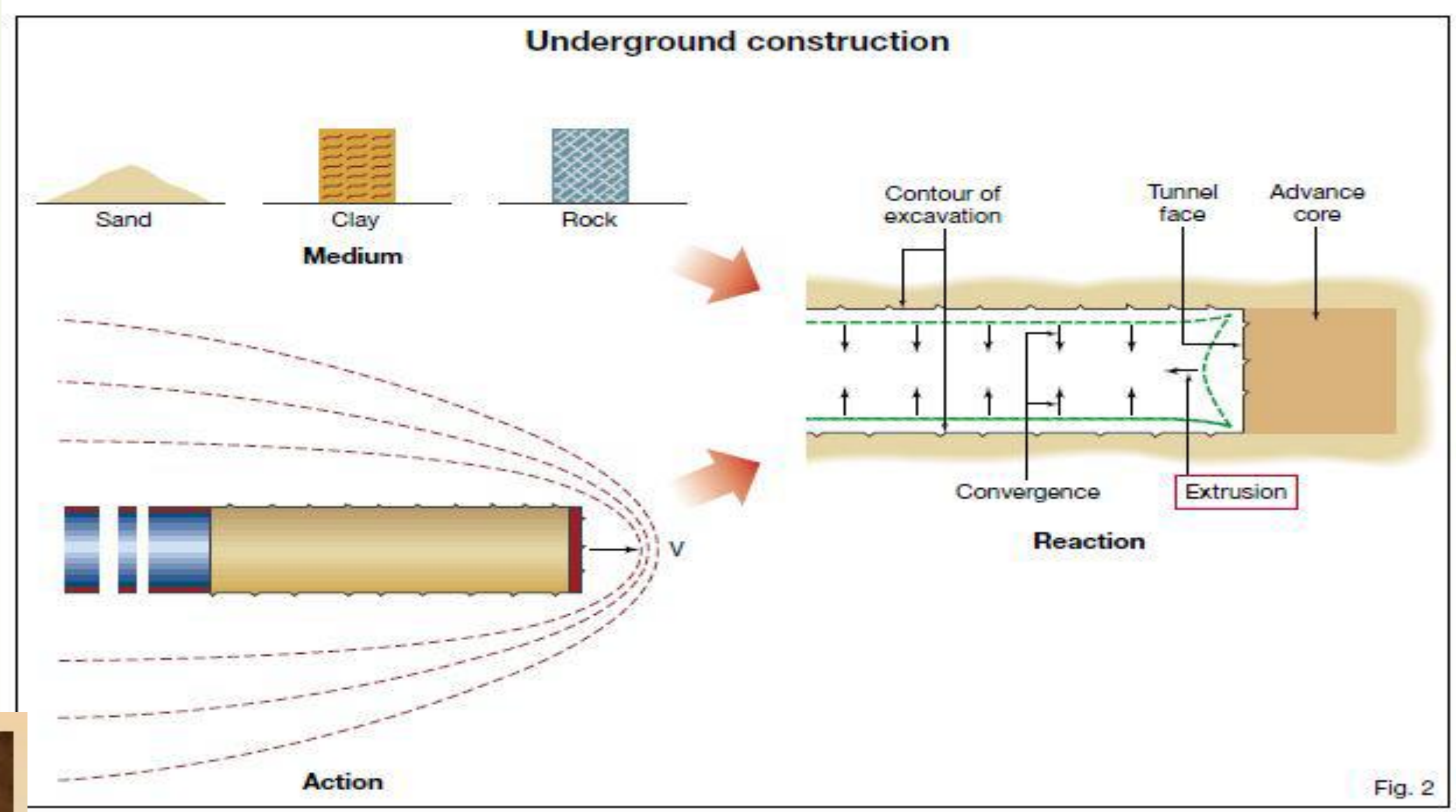
Developments layout

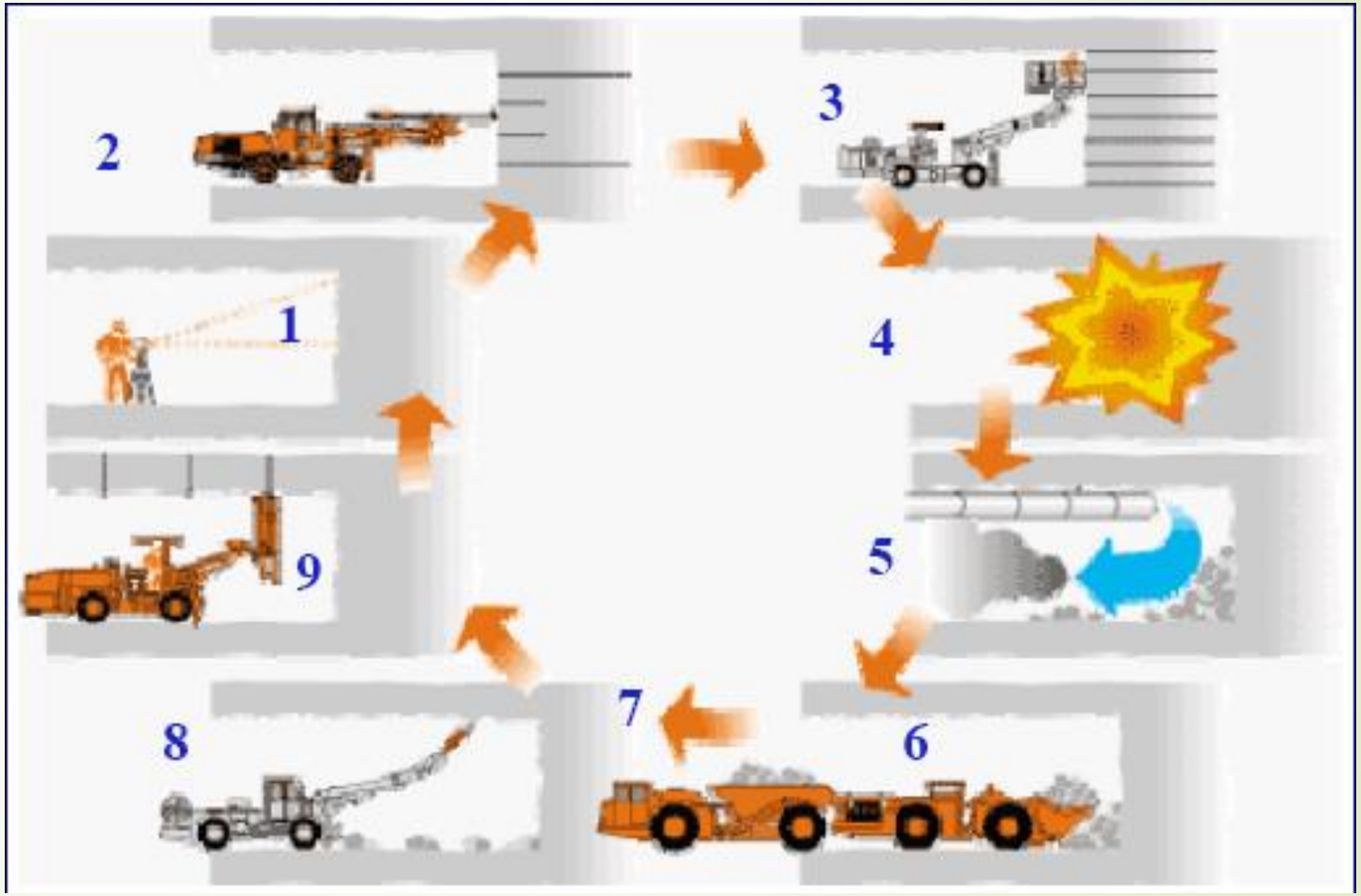


Drifting

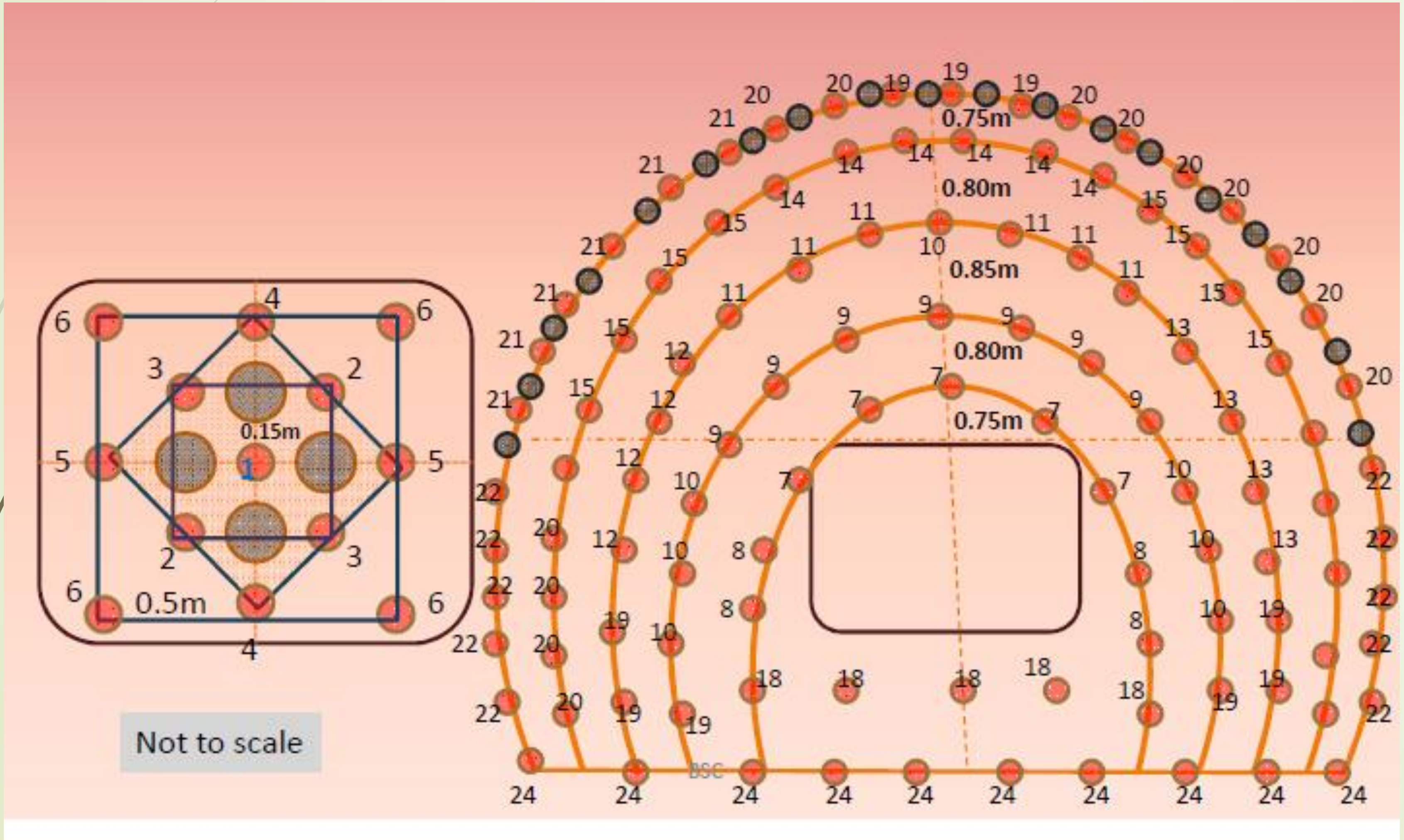

Underground construction
Sand
Clay
Rock
Medium
V
Action
Contour of excavation
Tunnel face
Advance core
Convergence
Extrusion
Reaction
Fig. 2

Drifting: Unit operations

1. Survey
2. Drilling
3. Charging
4. Blasting
5. Ventilation
6. Scaling
7. Mucking
8. Bolting



Drifting



Nomenclature of blast holes

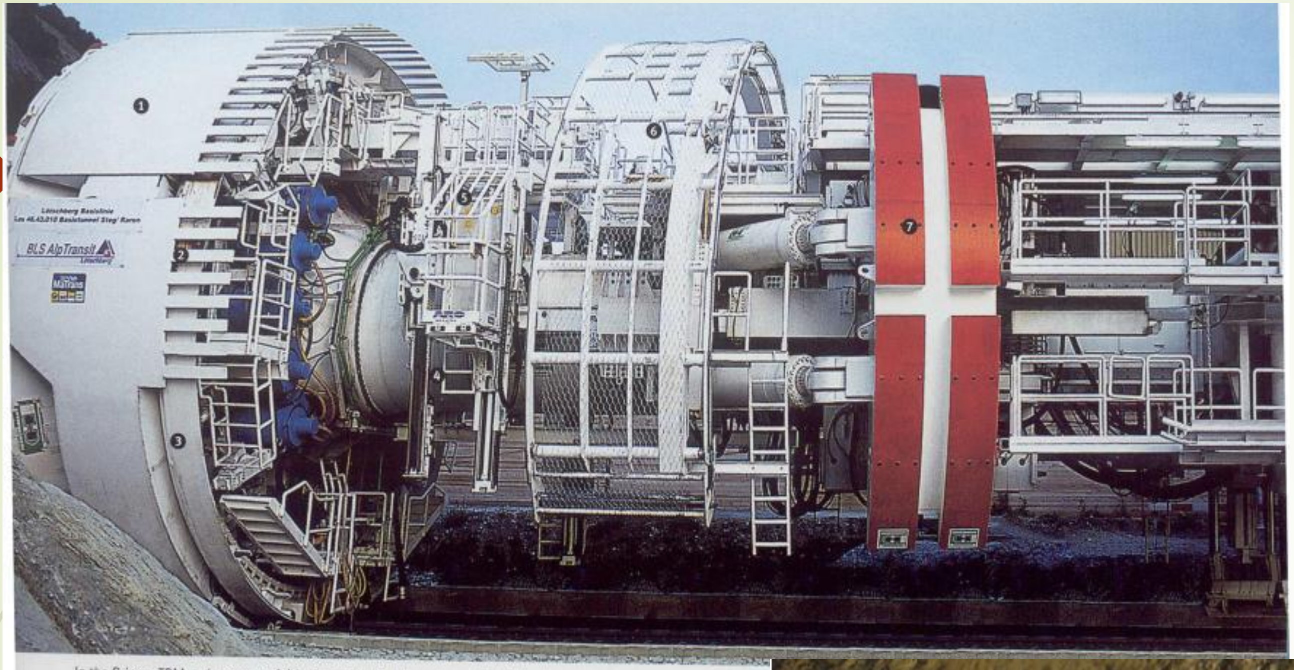
1.CutHoles

2.Easers

3.Lifters

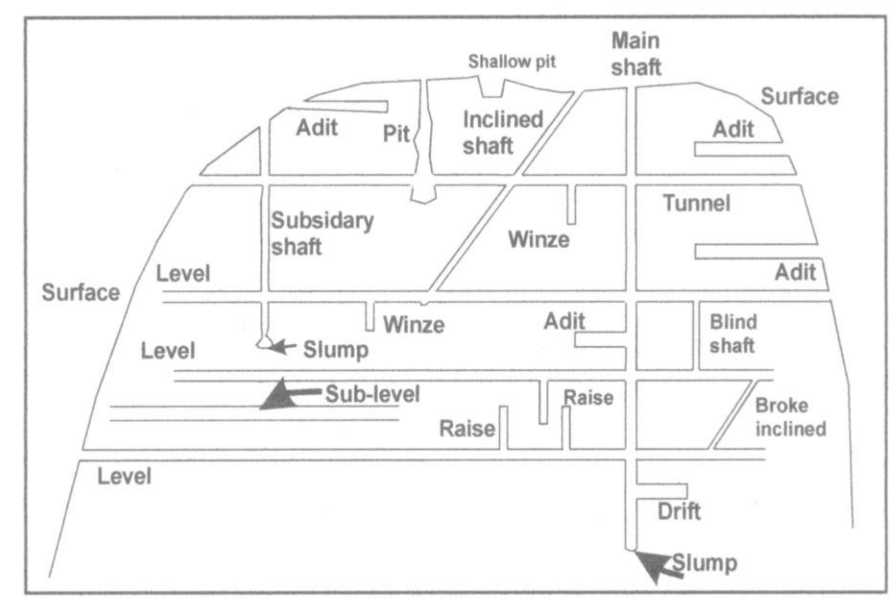
4.Stopingholes

5.Contourholes

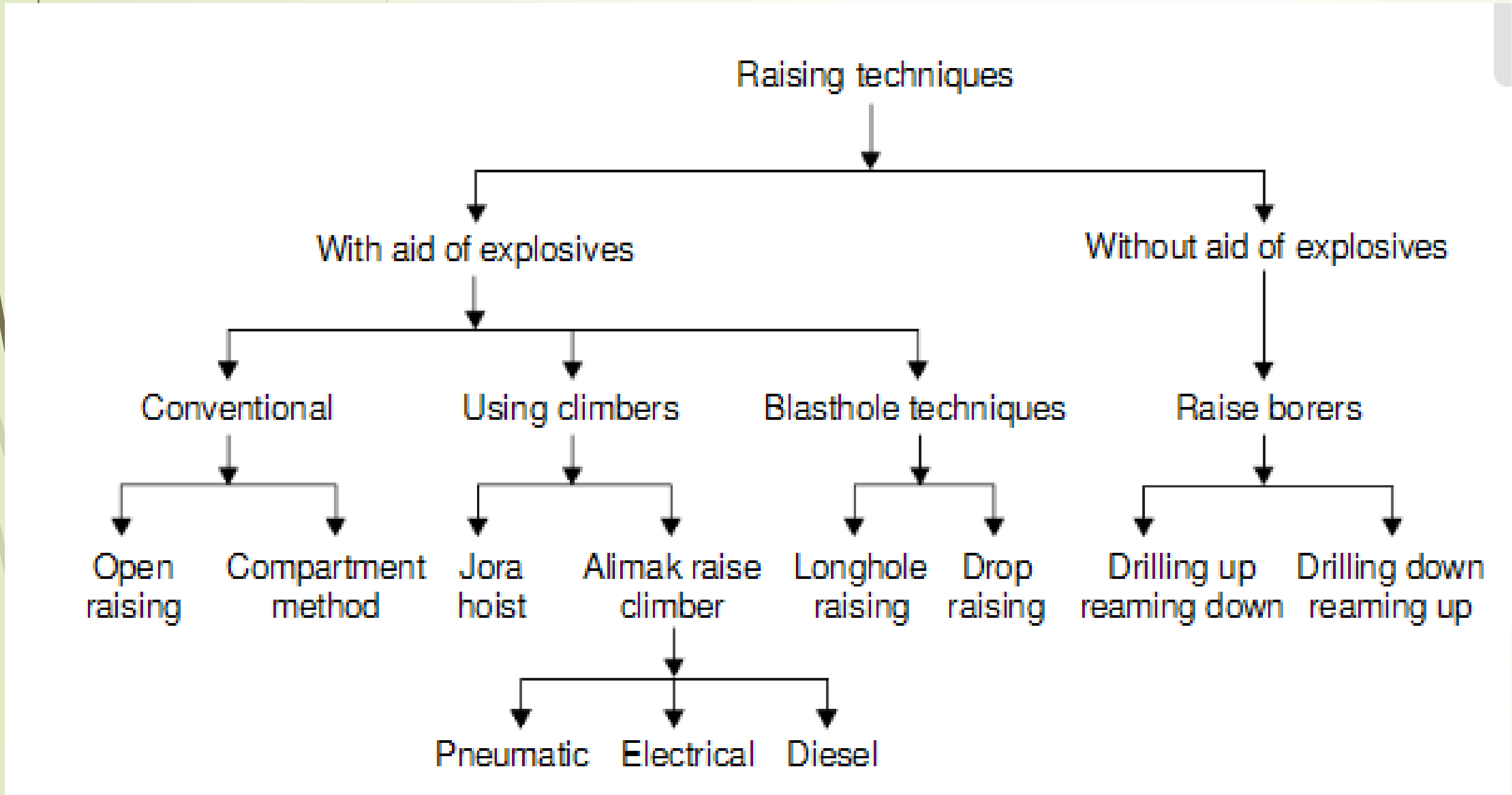

BLS AlpTransit

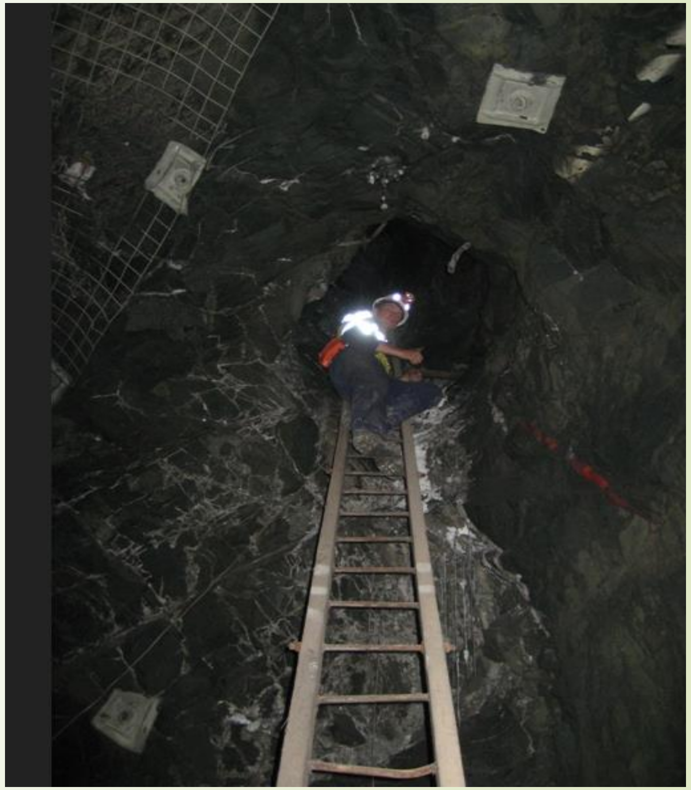
Terminology

- **Stopes** are openings from which ore is mined. They may be backfilled with cemented waste materials.
- **Drifts** are horizontal passageways used for access.
- **Ore passes** are sub-vertical chutes for movement of ore.
- **Declines or ramps** are spiral or inclined drifts.
- **Output**: In general for underground mines:
  - **small output mines** have output of < 4,000 tpd. Hauling is done on several levels, tonnage handled on each level is small, and light equipment is used.
  - **high output mines** have output of > 4,000 tpd. A main haulage level is used and all ore is dropped to that haulage level via ore passes.
- **Levels** include all the horizontal workings tributary to a **shaft** station. Ore excavated in a level is transported to the shaft to be hoisted to the surface.

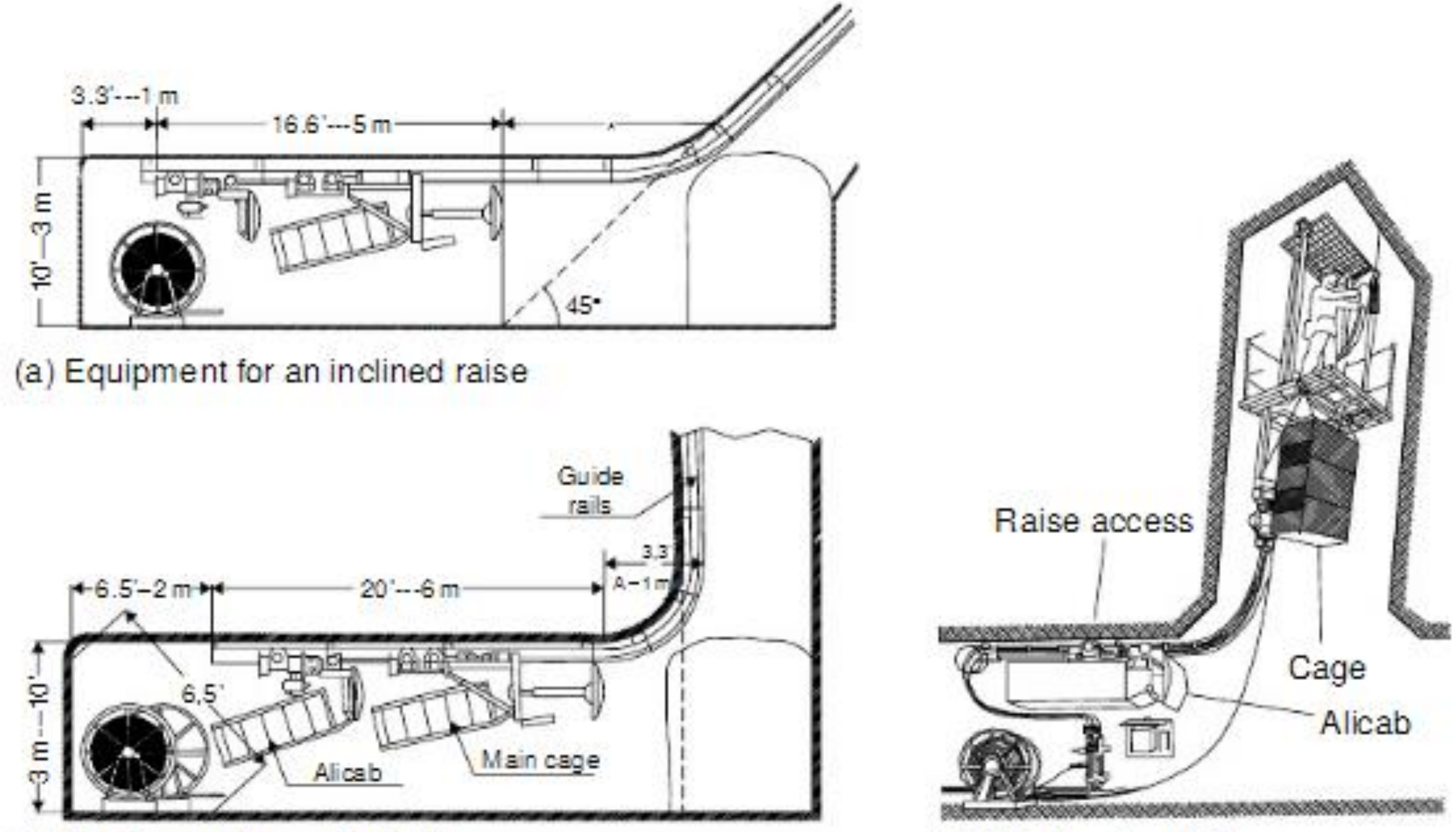


Raising


Raising techniques
With aid of explosives
Without aid of explosives
Conventional
Using climbers
Blasthole techniques
Raise borers
Open raising
Compartment method
Jora hoist
Alimak raise climber
Longhole raising
Drop raising
Drilling up reaming down
Drilling down reaming up
Pneumatic
Electrical
Diesel



Arrangement of raising in Alimak

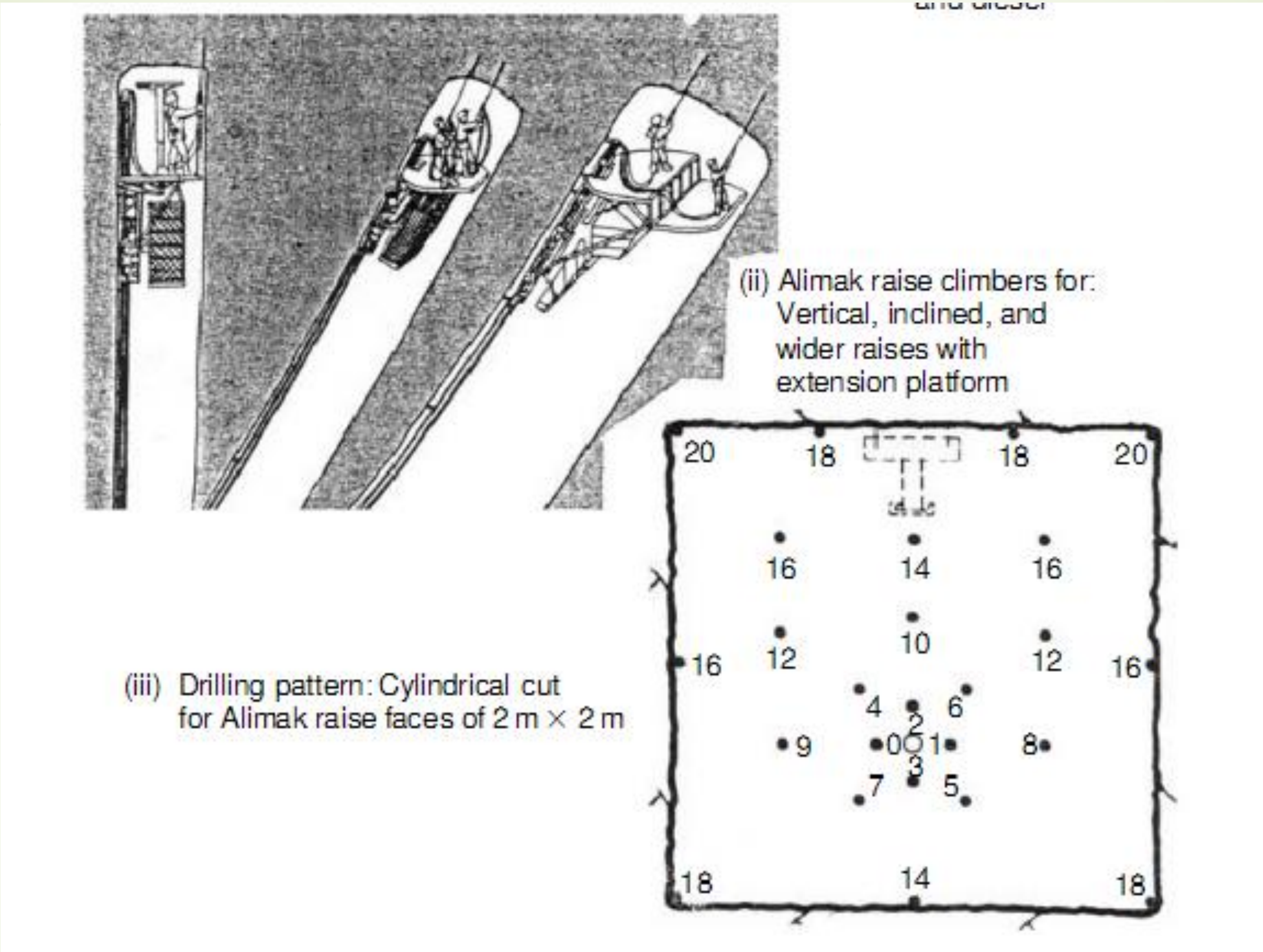


(a) Equipment for an inclined raise

(b) Equipment for vertical raise

(c) Alimak raise climber

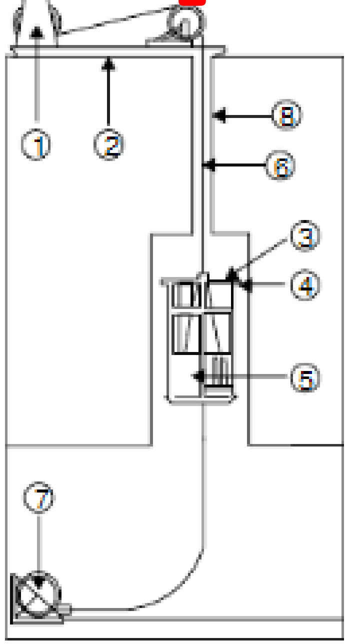
Drilling in Alimak raising



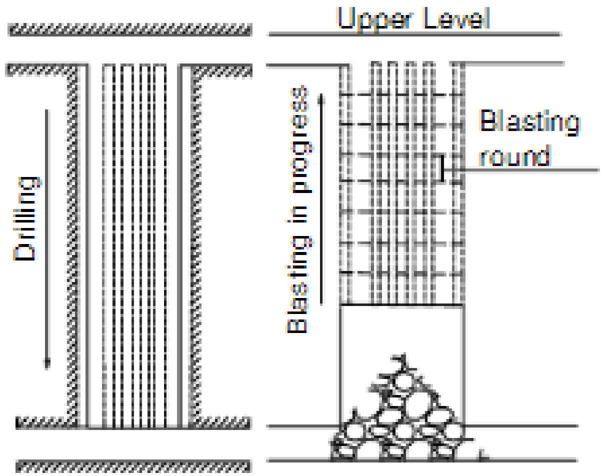
(ii) Alimak raise climbers for: Vertical, inclined, and wider raises with extension platform

(iii) Drilling pattern: Cylindrical cut for Alimak raise faces of 2 m × 2 m

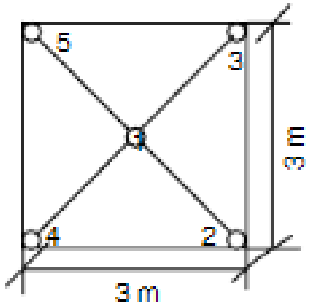
Various raising methods



(c) Raising with the Jora Lift



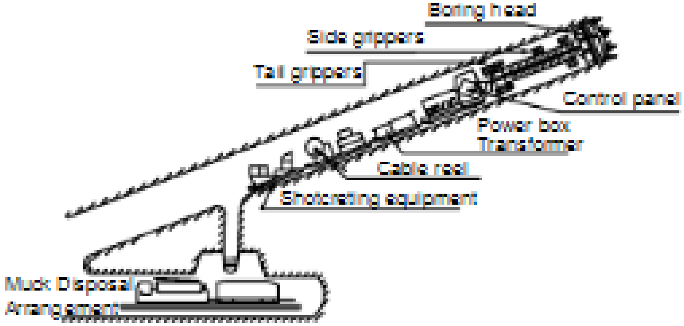
(d) Drilling with blasthole drifters hole length 10–30 m & dia. 50–100 mm. Long-hole raising



Drilling with large blasthole drifter. Hole length 25–100 m & dia. 100–200 mm.

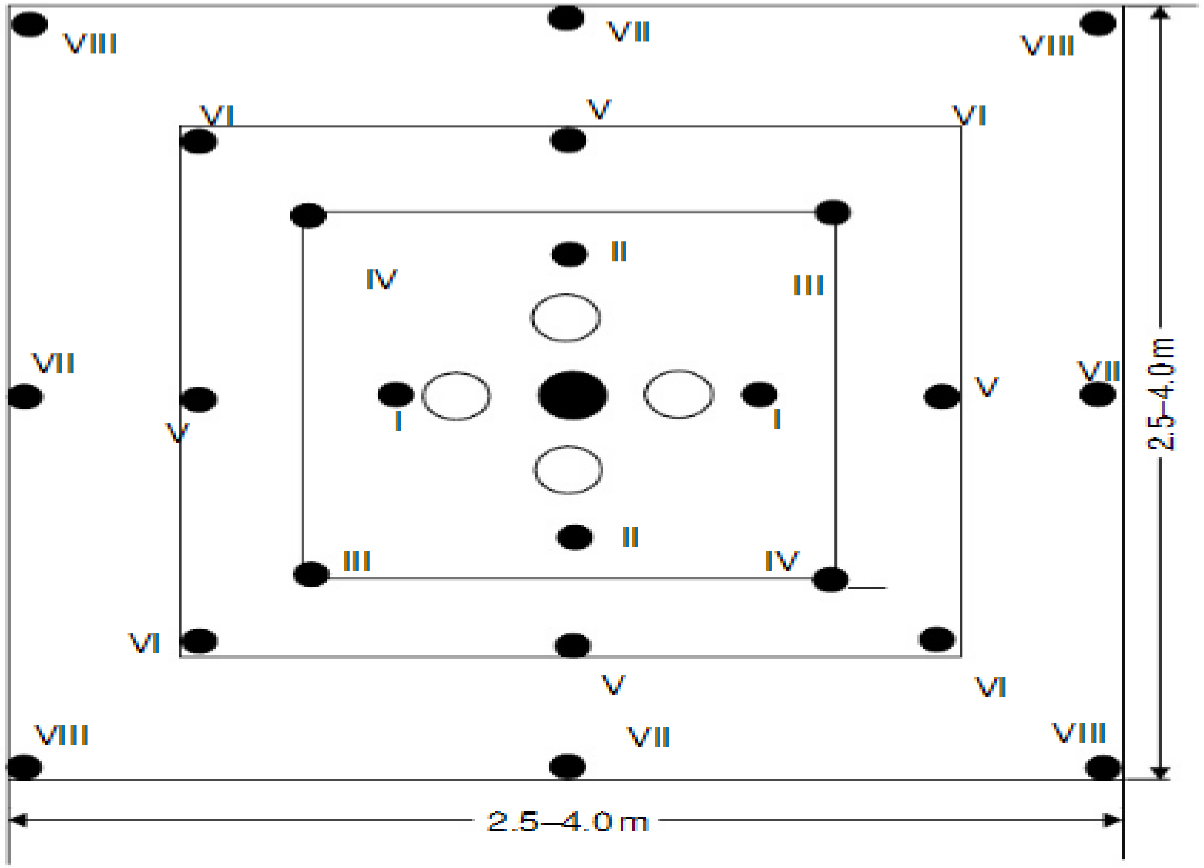
Delay no. (1 to 5)
165 mm – dia. hole

(e) Drop raising



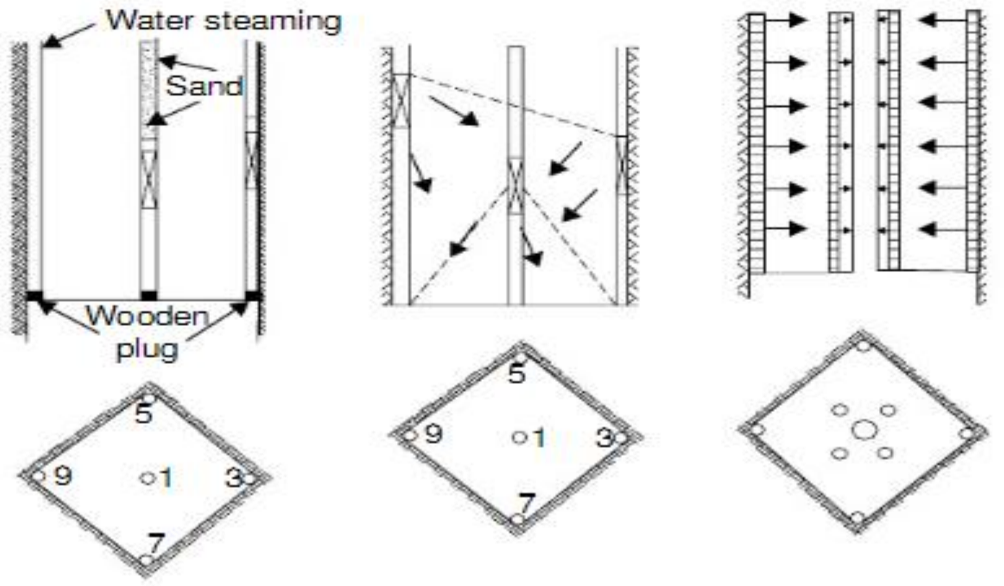
(f) Application of borer for driving inclines

Drilling pattern in long hole raising



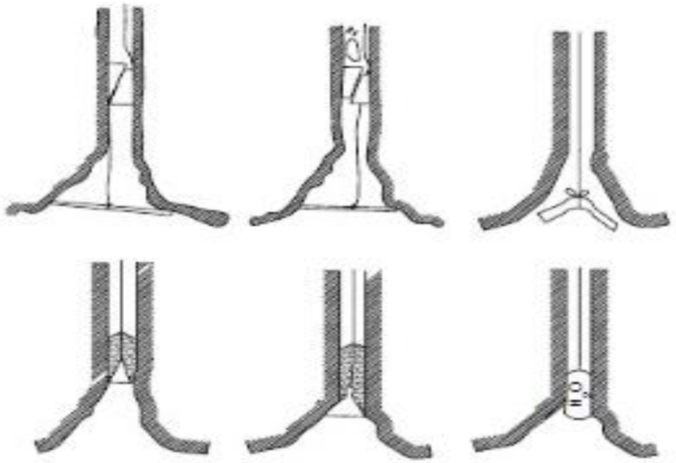
Legend
– Charged holes 50–55 mm dia.
– Uncharged holes 75–100 mm dia.
– Charged holes 75–100 mm dia.
– Roman letters indicate delay numbers

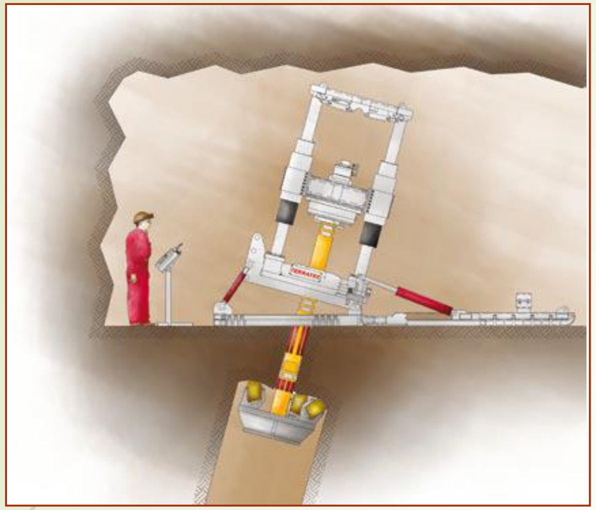
Drop raising method

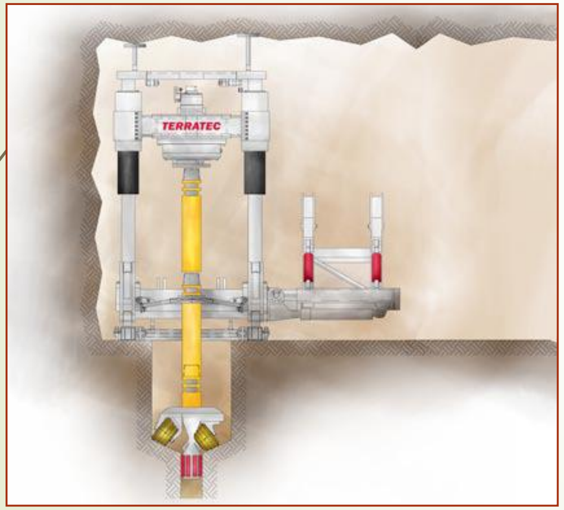


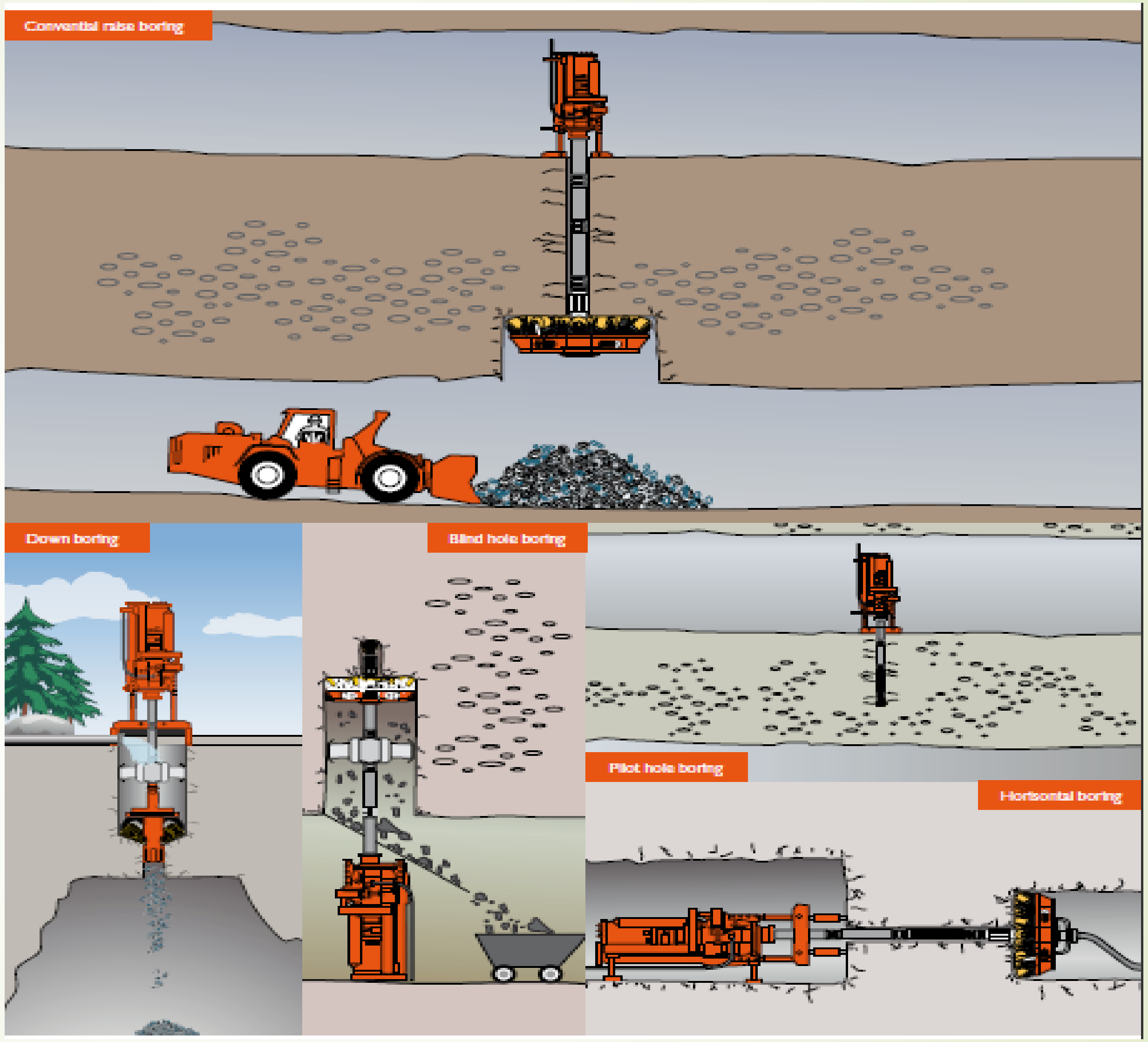
(a) VCR raising – some details

(b) Conventional cylindrical cut for large-hole raising





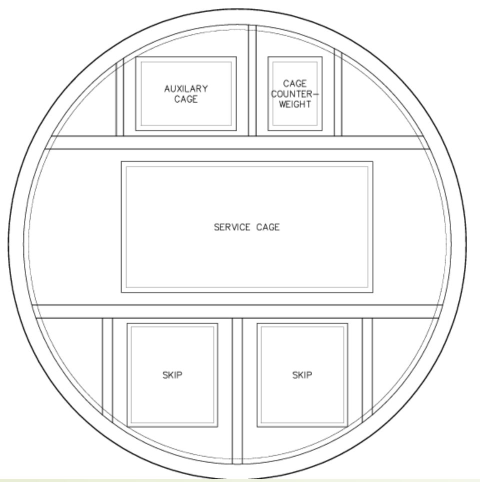


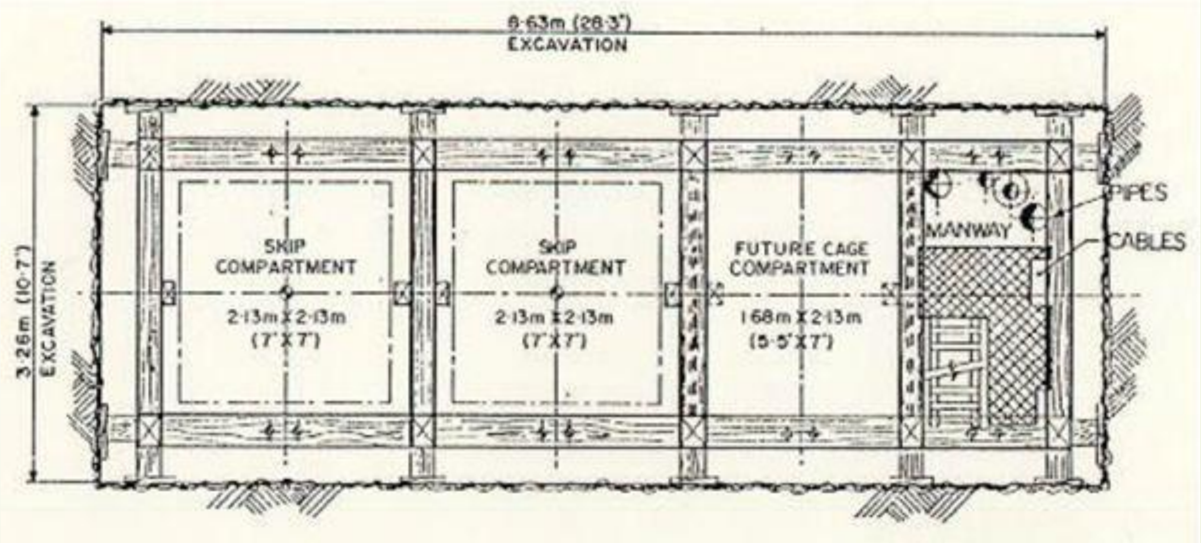


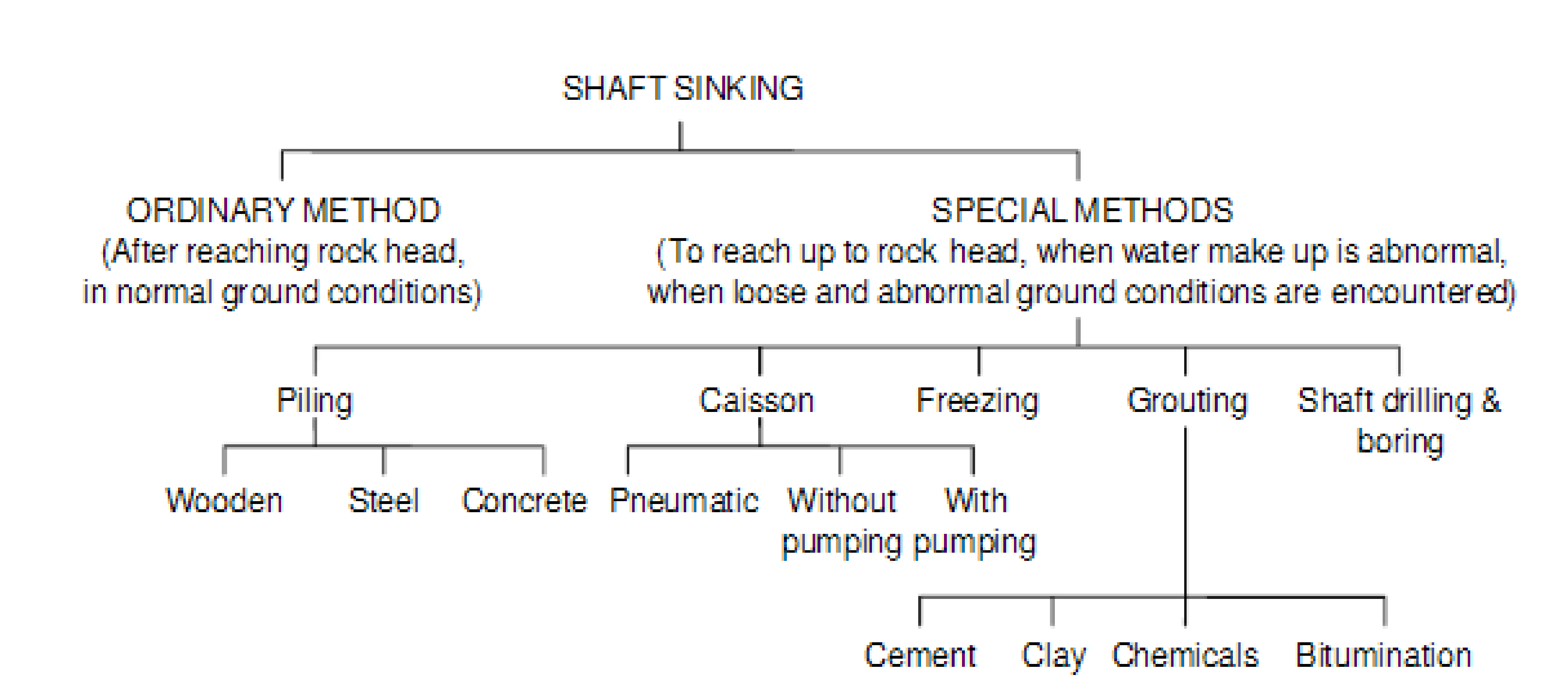
Raise-Boring

Shaft sinking

- To access an ore body
- To transport men and materials to and from underground workings
- For hoisting ore and waste from underground
- To serve as intake and return airways for the mine (ventilation)
- To provide a second egress as required by mining law
- Storage of nuclear waste






SHAFT SINKING
ORDINARY METHOD
(After reaching rock head, in normal ground conditions)
SPECIAL METHODS
(To reach up to rock head, when water make up is abnormal, when loose and abnormal ground conditions are encountered)
Piling
Caisson
Freezing
Grouting
Shaft drilling & boring
Wooden
Steel
Concrete
Pneumatic
Without pumping
With pumping
Cement
Clay
Chemicals
Bitumination

Factors Governing Choice of a Mining Method

- <u>**Shape and size of the deposit**</u>

Stock and Nests, **Columnar, Sheet**

- <u>**Deposit's contact with country rock**</u>

Massive ores have well defined contact with the country rock. In other cases, such as impregnation ores have no sharp contact with country rocks

Regular shape of ore body favours any of the mining methods but irregular and intricate shapes preclude uses of some systems or lower their efficiency.

Thickness of deposit

- Very thin deposits: thickness less than 0.7m.
- Thin deposits: thickness range 0.7–2m.
- Medium thick deposit: thickness range 2–5m.
- Thick deposit: thickness range 5–20m.
- Very thick deposit: thickness exceeding 20m.

Dip of the deposit

- Flat dipping: 0°to below 20°
- Inclined dipping: from 20°to below 50°
- Steeply inclined dipping: exceeding 50°

Influence of Dip

The dip of the deposit has the decisive influence on the selection of a mining method and positioning of a stoping face. Following are some of the important aspects regarding dip:

- Roof pressure decrease with increasing dip.
- Foothold of the workers deteriorates with the change of dip. Low dip provides firm foothold, further increase in dip does not give a firm foot hold and needs a safety belt. Working in moderate dips need platform to stand. For steep dip in over-hand stoping operations need an immediate filling of the worked out material, which may be sand, waste rock, mill tailings or accumulated broken ore to provide firm footing to the workers.
- With steeper dips and crumbly lode (ore body) there is a danger of falling the large pieces of ore, or sometimes caving of the whole face.
- Influence of dip during rock fragmentation: The process of rock fragmentation depends upon hardness, coherence and dip of the deposit. In overhand stoping gravity helps in varying degree according to dip of deposit i.e. most in vertical deposits, and least in the flat once.

- While mining vertical and steeply inclined deposits gravity together with weight of the rock helps the rock breaking process, whereas, this influence get diminished with decrease in dip.
- In steeply dipping weak strata working by underhand stoping i.e. from upper level towards the lower level becomes essential. But underhand stoping is associated with the danger of rolling boulders, roof fall or its caving.
- With increasing dip, the loading and transportation of the broken muck becomes easier.
- Working under the non-coherent deposit is dangerous, as collapse of roof may occur any time but steep dips and non-coherentness of the deposits help to design the caving methods of mining.
- All these factors indicate that to a considerable extent stoping methods change with dip

Physical and mechanical characteristics of the ore and the enclosing rocks

In underground mining the stability of ore and country rock is equally important as the rock strength. Stability is the ability of a massif, undermined (exposed) from beneath or sides, to resist caving for a certain period of time

Presence of geological disturbances and influence of the direction of cleats or partings

Geological disturbance means presence of any one of them or combination of more than one of structures such as fault, folds, joints, fissures, dykes etc. These structures usually require extra care in terms of strata stability, water seepage, gas leakage etc. making the mining process sometimes more tedious and slow.

Degree of mechanization and output required

For low output use of conventional machines, equipment and methods can be used but large output mines require bulk-mining methods deploying large sized fast moving equipment. A balance is required to be made between the funds available to invest and the output rates to achieve optimum results.

Ore grade and its distribution, and value of the product

Ore value is the focal point while selecting a mining method. Based on the market value of the useful contents in the ore, cut-off grade of any mineral can be decided. The cut-off grade further decides ore reserves, as mineral reserves in any deposit below cut-off grade are considered waste and the one at and above cut-off grade as ore reserves. In high valued ores, costly mining methods, even at great depths, can be applied. Based on this logic, one will find that most of the gold deposits being mined are currently, are at great depths.

Depth of the deposit

Deepening of workings below 600–800m is often accompanied by a considerable rise in rock pressure, thus impeding the use of some systems. In addition, abrupt inrush of ore or rock burst from the stressed pillars is a phenomenon that has been encountered at very deep horizons. This warrants change in stoping methods.

Presence of water

Presence of gases

Ore & Country rocks susceptibility to caking and oxidation

Cut-off grade, minable ore tonnage, and average grade

Output, Rate of progress, recovery, Easy access within stope, Ore handling, etc

Environmental parameters

Mining/safety/regulatory parameters

Economic parameters

Criteria for Selection of Stoping Method

Factors	Considerations
Geological parameters	1. Lithology and structural details 2. Groundwater 3. genesis
Ore body characteristics	1. Dip amount and direction 2. Shape and size 3. Grade vis-à-vis tonnage and their spatial distribution 4. Physico-mechanical and physico-chemical characteristics
Host rock characteristics	Physico-mechanical and physico-chemical characteristics
Cut-off grade, minable ore tonnage, and average grade	
Economic parameters	1. Capital and operating costs 2. Environmental pollution control costs 3. Mine closure costs 4. Market demand and mineral price
Environmental parameters	1. Land degradation potential 2. Noise and air pollution potential 3. Surface water contamination potential 4. Groundwater contamination potential
Mining/safety/regulatory parameters	1. Degree of mechanization and manpower requirement 2. Ventilation and refrigeration requirement 3. Support requirement 4. Dust, noise and gas control requirement 5. Illumination requirement

Different U/G Metal Mining Methods

- Self Supported Methods
  - Room and Pillar Mining
  - Stope and Pillar Mining
  - Shrinkage Stoping
  - Sublevel Stoping
  - VCR Mining
- Supported (Cut and Fill Stoping) Methods
  - Overhand Cut and Fill Stoping
  - Square Set Stoping
  - Post Pillar Stoping
  - Undercut and Fill Stoping
  - Drift and Fill Stoping
- Caving Methods
  - Sublevel Caving
  - Block Caving
- Novel and Innovative Methods

i) *Unsupported underground mining*

a) Room (Bord)-and-Pillar mining (or continuous mining) method

- It is the most common unsupported method, designed and used *primarily for mining flat-lying seams or gently dipping bedded ore deposits of limited thickness (like coal, trona, limestone, and salt).*
- It is preferred to apply for sedimentary deposits (such as shales, limestone, dolomite or sandstone) containing copper, lead, coal seams, phosphate layers, and evaporate (salt and potash) layers.
- Room and pillar methods are well adapted to mechanization, and are used in tabular orebodies such as *coal, potash, phosphate, salt, oil shale, and bedded uranium ores.*
- Pillars are left in place in a regular pattern *while the rooms are mined out.*
- Support of the roof is provided by natural pillars of the mineral that are left standing in a systematic pattern.
- The mining cavity is supported (kept open) by the strength of remnants (pillars) of the orebody that are left un-mined.

- Room-and-pillar mining method has a low recovery rate (a large percentage of ore remains in place underground).
- In many room and pillar mines, the pillars are taken out starting at the farthest point from the stope access, allowing the roof to collapse and fill in the stope. This allows for greater recovery as less ore is left behind in pillars.
- It is an advantageous mining method for shallow orebodies –as a means of preventing surface subsidence. Historic, ultra-shallow underground coal mines (<30 m) nevertheless are characterized by surface subsidence in the areas between pillars.
- Pillars are sometimes mined on retreat from a working area, inducing closure and caving of these working panels, and raising the risk of surface subsidence.

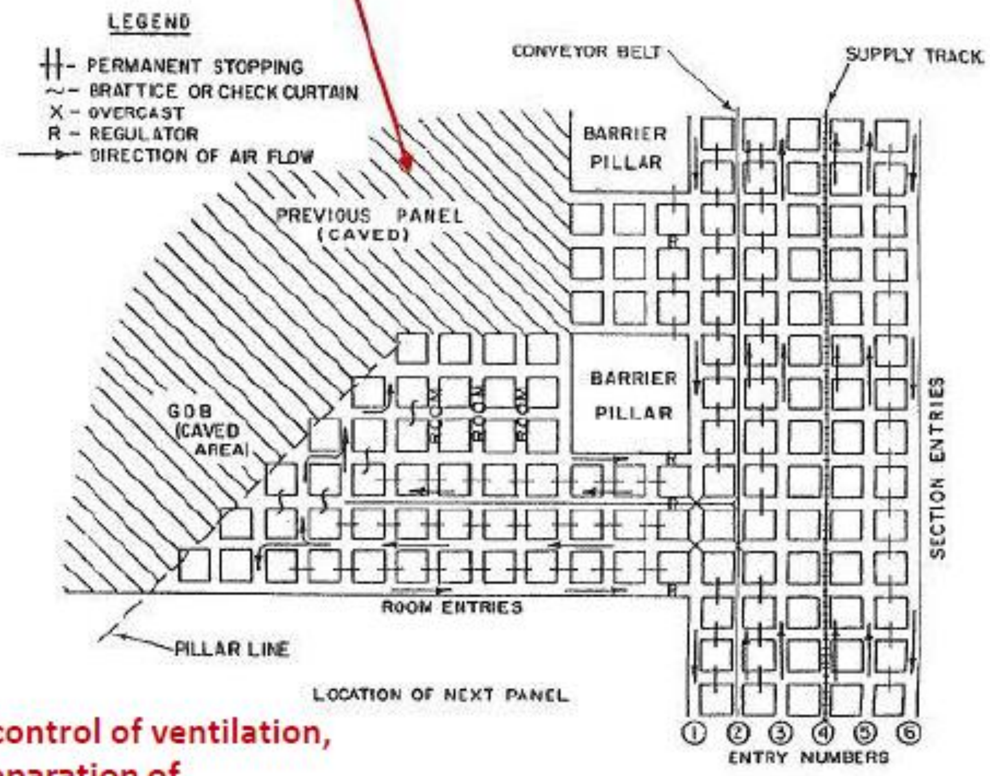
Conditions & main elements:

Following are the suitable conditions for this method:

- Ore strength: weak to moderate (its variants can mine out strong ore bodies also).
- Rock strength: moderate to strong.
- Deposit's shape: tabular.
- Deposit's dip: low (usually below 15°, but its variants can mine out ore bodies up to 40°dip).
- Size and thickness: large extent, thickness below 5m, if more benching will be required.
- Ore grade: moderate.
- Depth: Shallow to moderate (less than 500m).

Room (Bord)-and-Pillar Layout

Pillars have been mined-out in this area



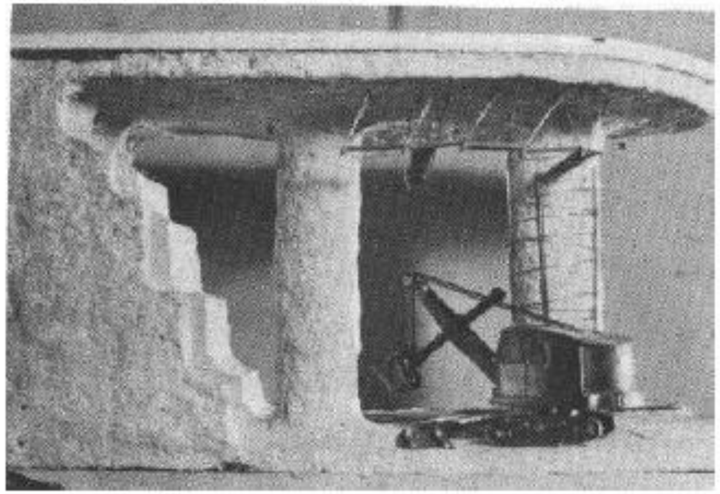
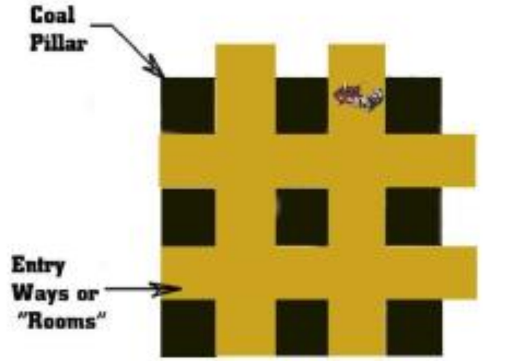


Figure shows Room and Pillar Mining



Note the control of ventilation, i.e., the separation of contaminated (used) and uncontaminated (fresh) air using a variety of devices.

Figure from Hartman and Mutmansky, 2002.

Underground mining: room-and-pillar mining of thick seams –"benching"

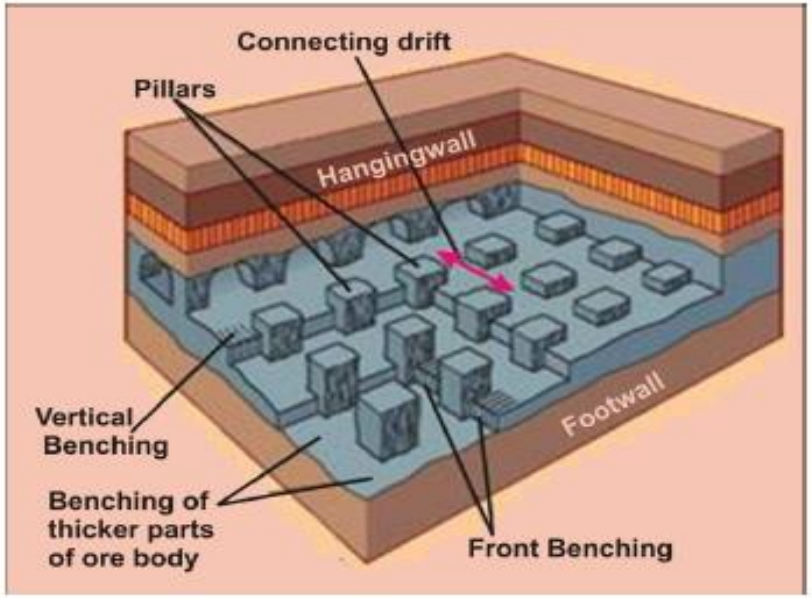
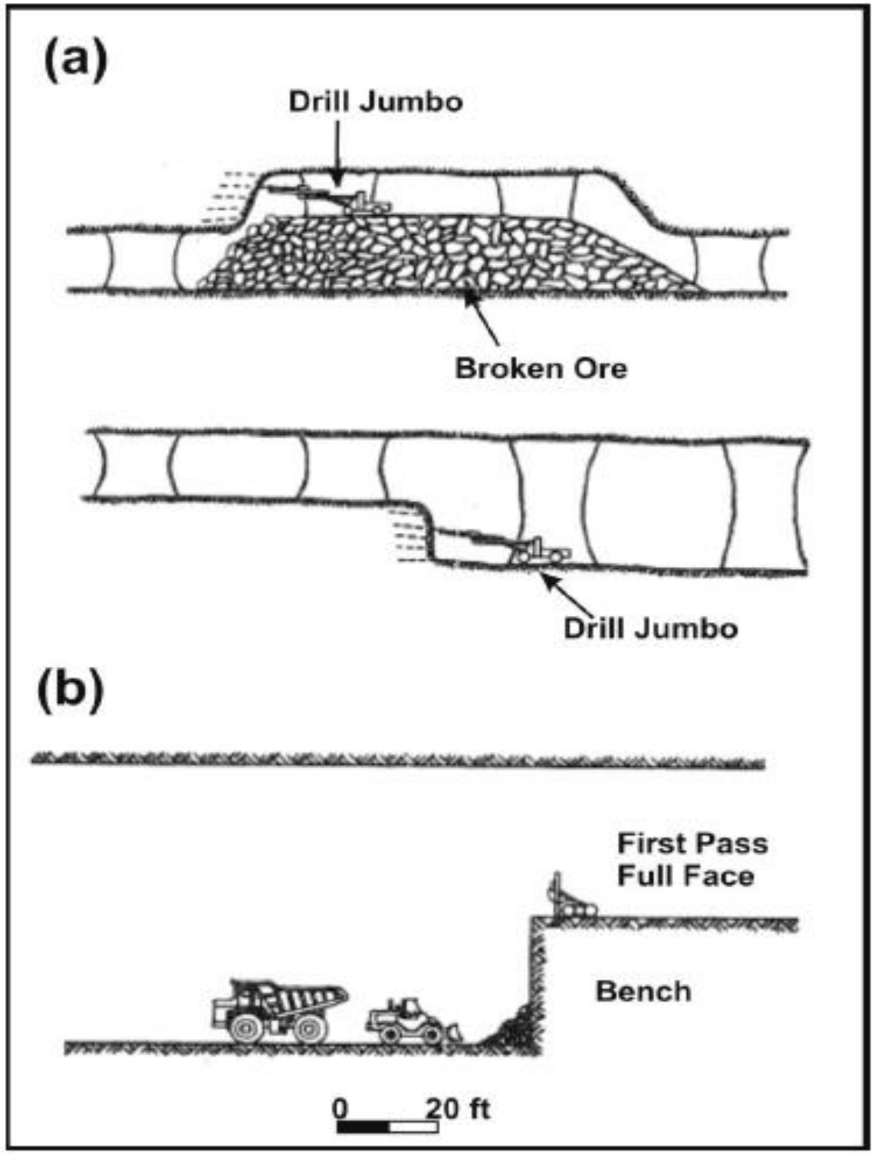


Figure shows Room and Pillar is designed for mining flat, bedded deposits of limited thickness.

Different approaches allow either the top or bottom part of the seam to be mined out first.

Note: the ***"hangingwall"*** is above the mining cavity, and the ***"footwall"*** is below it.



Figures from Hartman and Mutmansky, 2002.

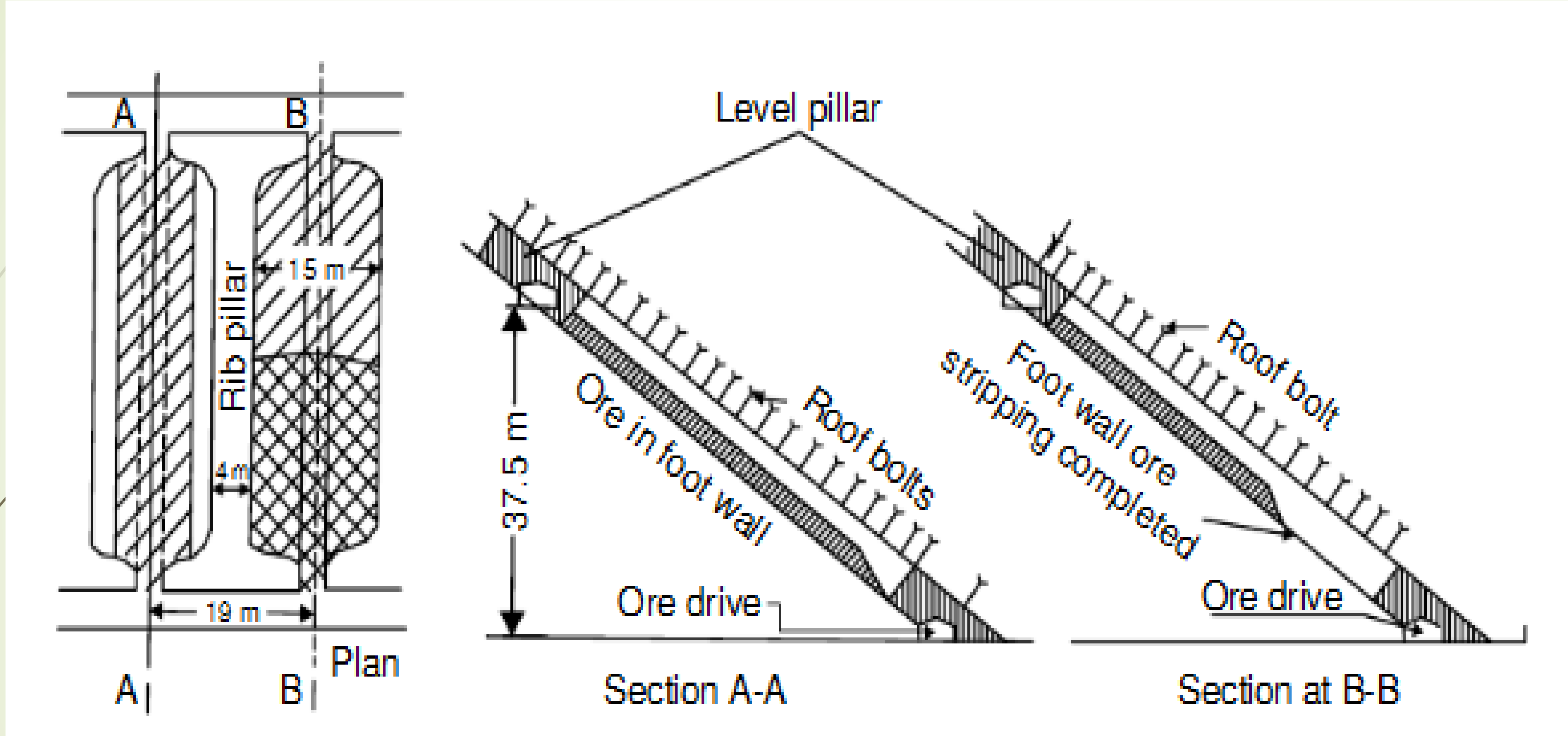


Figure: Layout of a stope and pillar stope to mine out an inclined 5–6m thick deposit in two slices

Advantages

- Productivity: Moderate to high. There is a very good scope for mechanization in this method. But its variant bord and pillar is labour intensive.
- Range of OMS: Overall OMS 15–30t/shift/man and face OMS 30–70t/shift/man.
- Cost: Low to moderate (30% relative cost, when compared with Square set stoping which is the costliest method – rating given 100%).
- Production rate: Moderate to high
- Dilution: maximum up to 10–20%, Selective mining possible. Sometimes lean patches of ore body are left as pillars.
- Flexibility: unit operations can be carried simultaneously. Also change in layout, sets of equipment and number of working faces possible.
- Development cost and time: most of the development work is in ore; as such during development production up to 20–25% can be obtained. Quantum of development work is also not high, as such; bringing the stopes quickly into the regular production phase is possible.

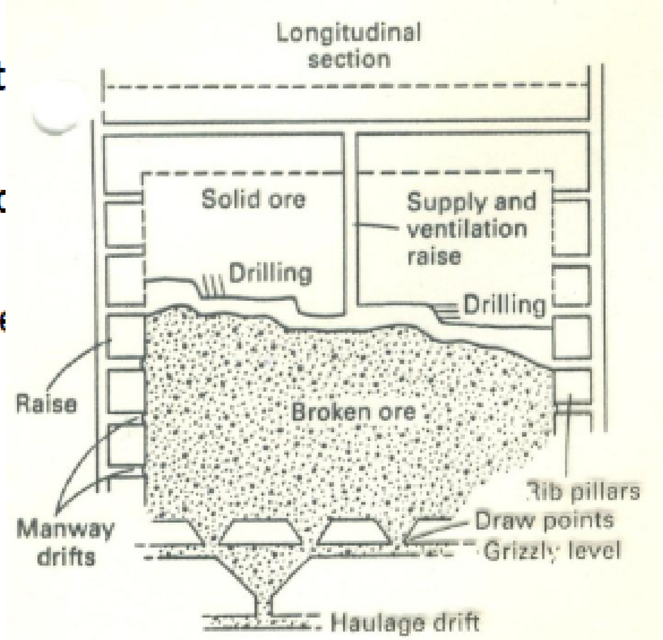
Limitations

- During stoping use of temporary supports is necessary. This lowers productivity and increases costs.
- Recovery: without pillar extraction 40–50% and with pillar extraction 70–90%.
- Skill: requires special skill during development as well as stoping operations.
- Capital required: sufficient.
- Safety: working under exposed roof is a potential source of accidents. In deposits thicker than 3m working in benches under high roofs is dangerous.
- Ventilation in large open stopes is sluggish

b) Shrinkage Stoping

- It is a **short hole mining method** which is *suitable for steeply dipping orebodies.*
- Mining progresses upward, with horizontal slices of ore being blasted along the length of the stope. A portion of the broken ore is allowed to accumulate in the stope to provide a working plat form for the miners and is there after withdrawn from the stope through chutes.
- The method is ***similar*** to cut and fill mining with the *exception that after being blasted, broken ore is left in the stope where it is used to support the surrounding rock and as a platform from which to work.*
- Only enough ore is removed from the stope to allow for drilling and blasting the next slice.
- The stope is emptied when all of the ore has been blast
- Although it is very selective and allows for low dilution, in the stope until mining is completed there is a delayec investments.[3]
- It is more suitable than sublevel stoping for stronger ore

Note: *Sublevel stoping* differs from shrinkage stoping by providing sublevels from which vertical slices are blasted. In this manner, the stope is mined horizontally from one end to the other.



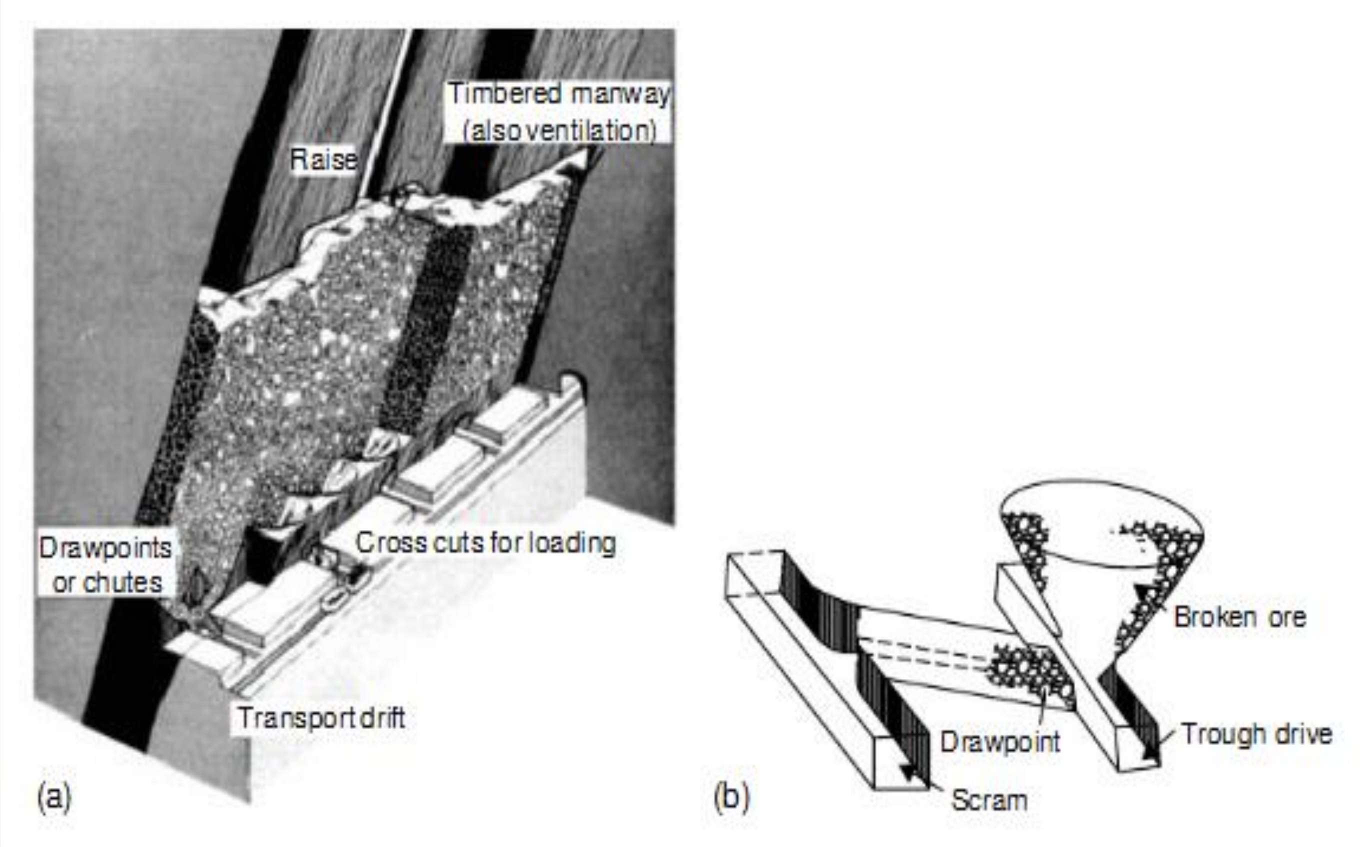


Figure: Pictorial view – shrinkage stoping

Following are the suitable conditions, in general, for the application of shrinkage stoping:

- Ore strength: strong (without caking, oxidizing & spontaneous heating).
- Rock strength: strong to fairly strong.
- Deposits shape: tabular or lenticular, regular dip and boundaries.
- Deposit's dip: steep, (preferably 60–90°).
- Size and thickness: Large extent, narrow to moderate thickness but not below 1mand up to 30m (up to 15m is common, In fact rarely for the thick ore bodies).
- Ore grade: fairly uniform and high.

- Depth: practiced up to 750m.

Unsupported Underground Mining Methods		
Factor	**Room and Pillar Mining**	**Shrinkage Stoping**
Ore strength	Weak \ Moderate	Strong
Rock strength	Moderate \ Strong	Strong
Deposit shape	Tabular	Tabular \ lenticular
Deposit dip	Low \ Flat	Fairly steep
Deposit size	Large \ Thin	Thin \ Moderate.
Ore grade	Moderate	Moderate
Ore uniformity	Uniform	Uniform
Depth	Shallow \ Moderate	Moderate

In ***"unsupported"*** mining, the mine-workings are supported temporarily only for as long as needed to keep the active face open to mining. ***After mining, the support*** (e.g. hydraulic props or wood packs) is removed (or becomes crushed), and the mining cavities close up under the pressure of the overburden material. ***The cavity closure is either partial, for shallow mining, or complete, for deep level mining***.

While unsupported mining is advantageous in that it ***maximizes ore recovery*** (as little ore as possible is left behind) the method comes with ***significant problems***:

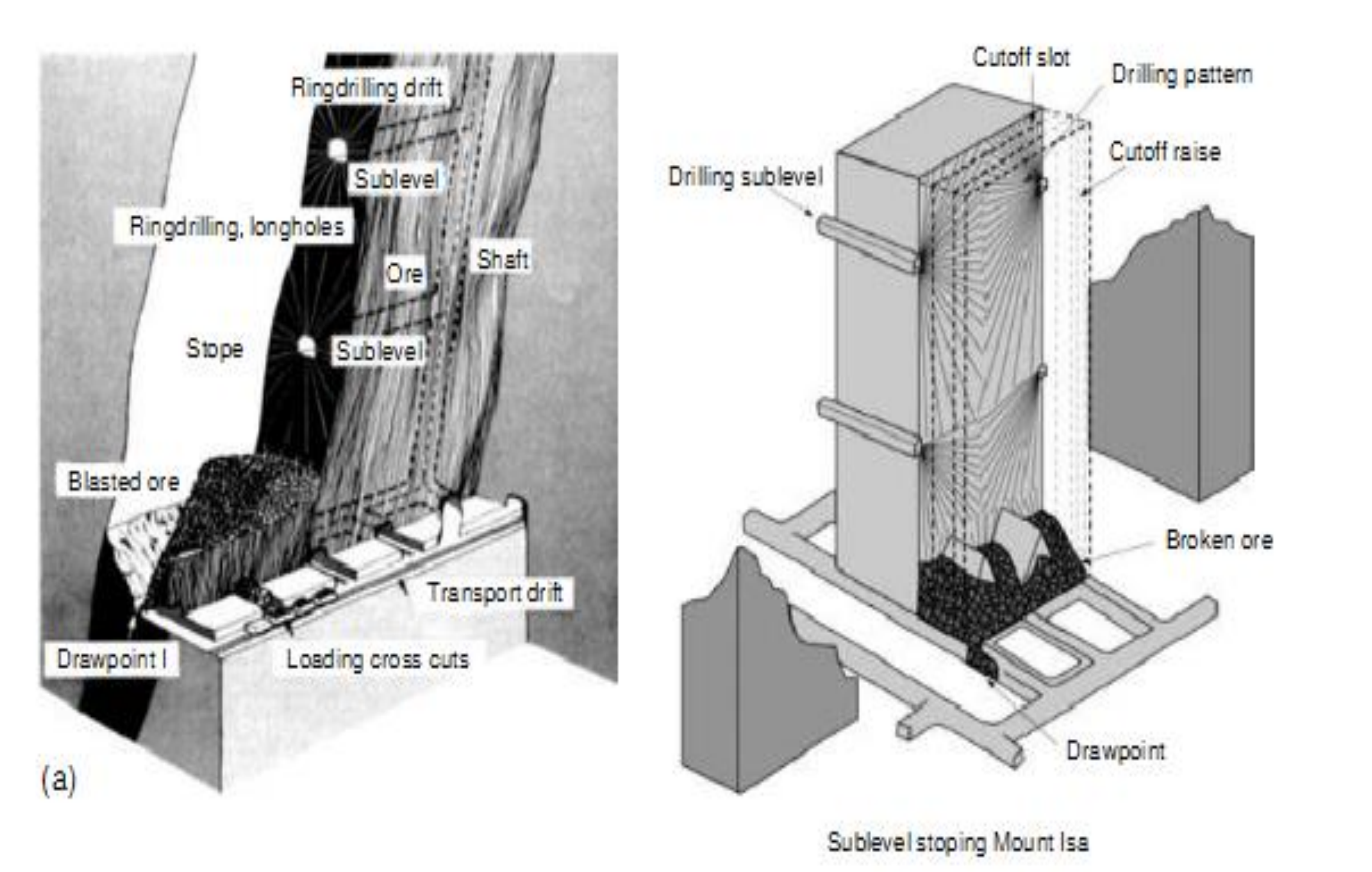
- Surface subsidence in the case of shallow mines.
- Rockbursts underground, causing injury and death in deep level mines.

Open Stoping Method – Sublevel Stoping

In this method the ore body is vertically divided into levels, and between two levels the stopes of convenient size are formed. A rib pillar left in between them separates two adjacent stopes. Leaving a crown pillar at the top of the stope protects the level above, whereas lower level is used as haulage level to gather the ore from the stopes. Vertically the stope is divided into a number of horizons by suitably positioned 'drill drives', called sublevels, and hence the name 'sublevel stoping'

Sublevel stoping can be classified as under:

- Sublevel stoping with benching – with the use of conventional drills.
- Blast hole stoping – with the use of blast hole drills. (VCRM)
- large blast hole stoping – with the use of large dia. drills such as DTH drills.



Pictorial view – sublevel stoping.

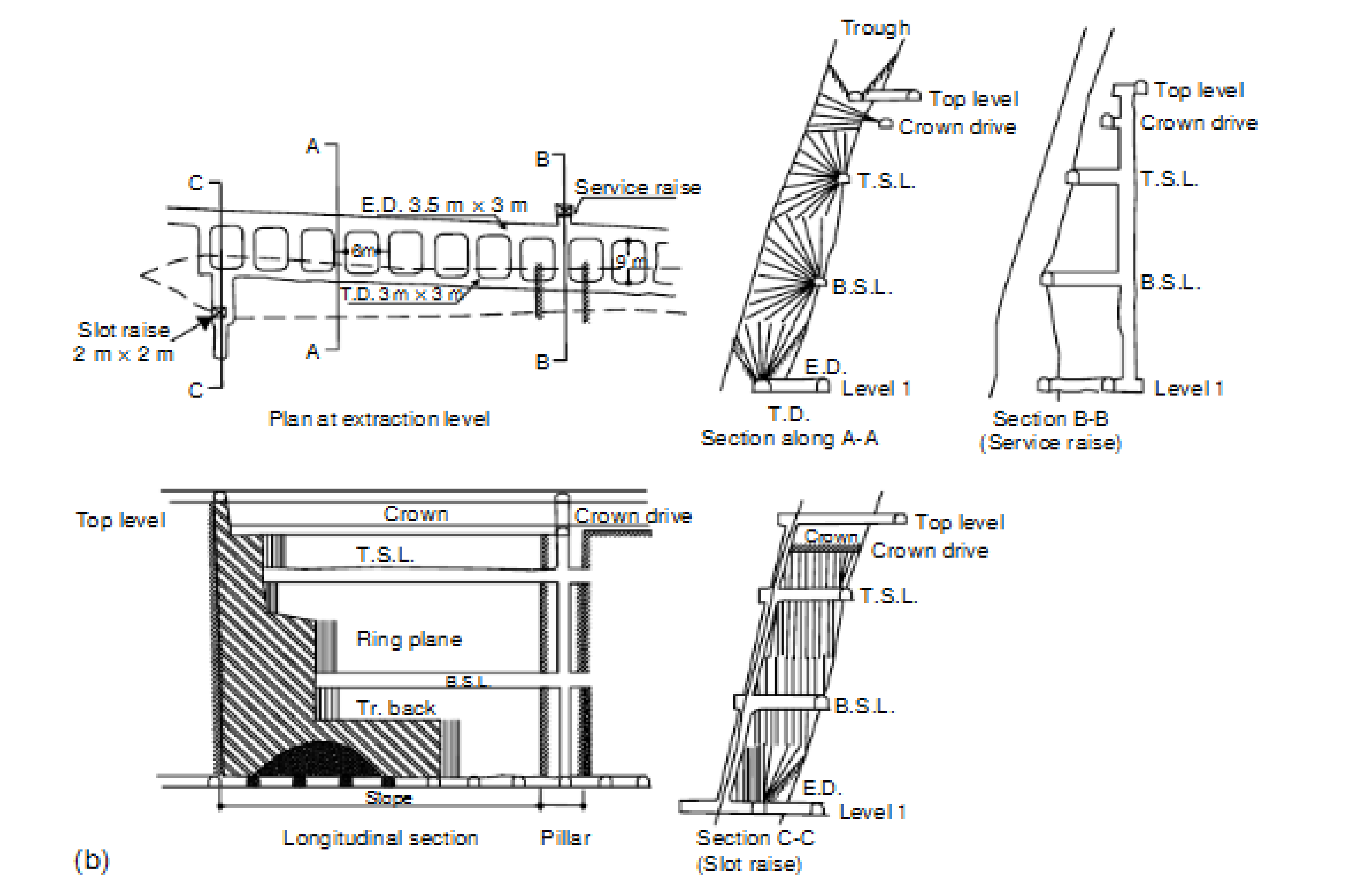
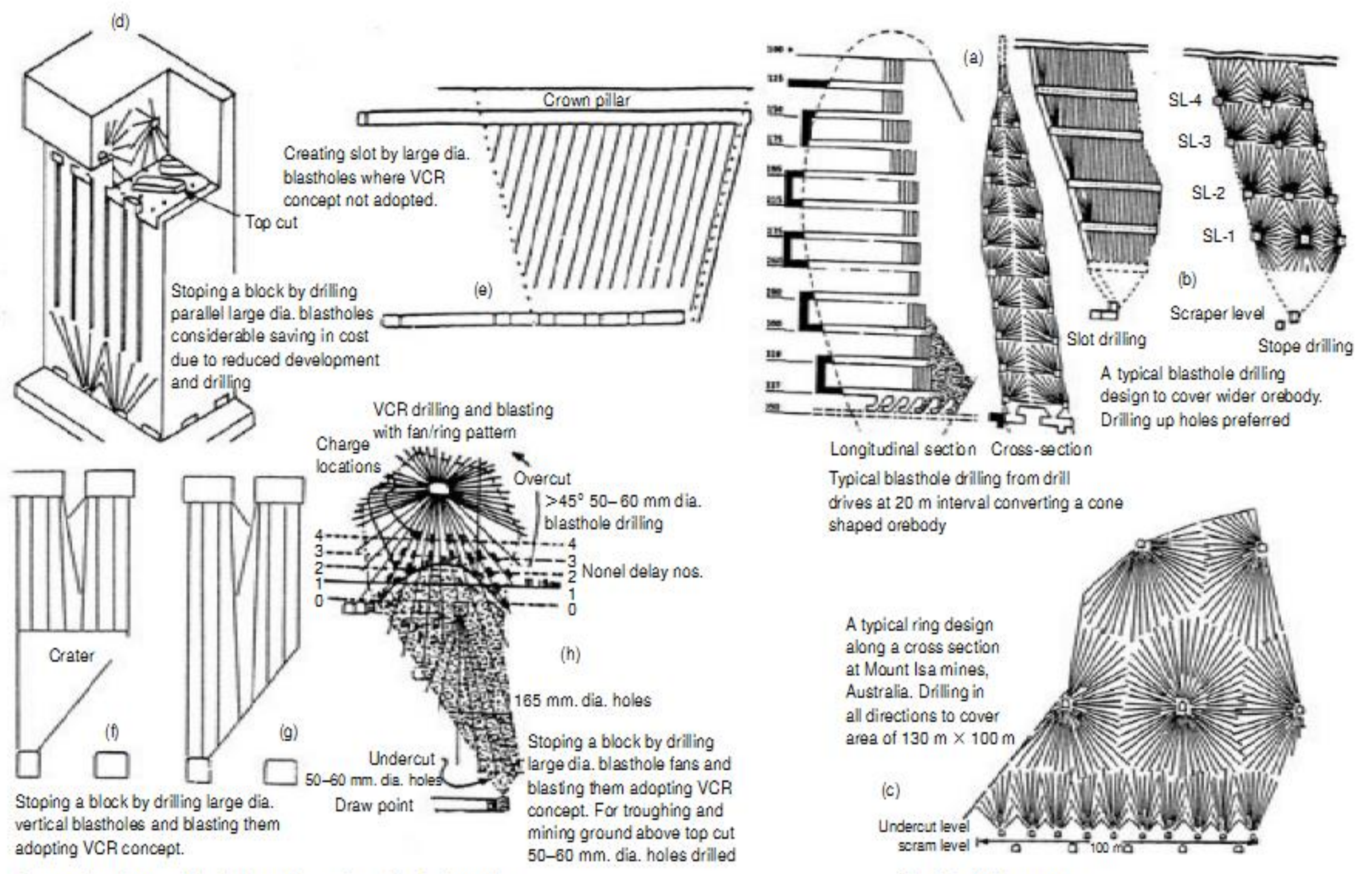


Figure: Longitudinal sublevel stoping



A comparison between blasthole stoping and large blasthole stoping

Blasthole drilling patterns for blasthole/sublevel stoping

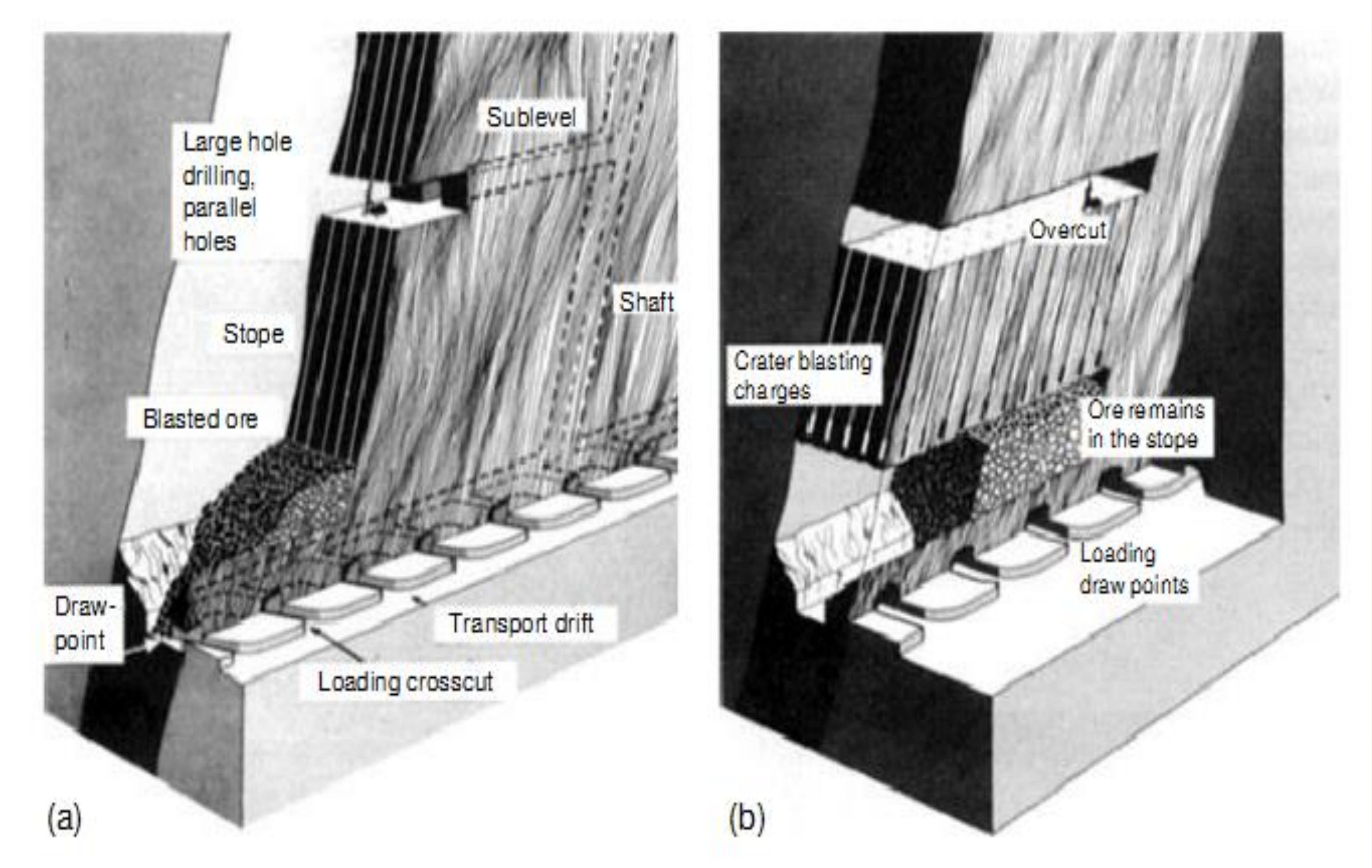


Figure : Big-blast hole/DTH stoping – pictorial view. (b) VCR stoping – pictorial view

Open Stoping Method – Sublevel Stoping

Suitable conditions in general, for the application of sublevel stoping:

- Ore strength: moderate to strong.
- Rock strength: fairly strong to strong.
- Deposits shape: tabular or lenticular, regular dip and boundaries
- Deposit's dip: steep, (preferably 60–90°).
- Size and thickness: large extent, thickness not below 5m and up to 30m or more.
- Ore grade: fairly uniform.
- Depth: practiced up to 1km. or even more.

Applications: Most popular hard rock mining method in underground metal mines to mine variety of ores. This includes copper, lead, zinc, iron, sulphur, nickel, gold and many others. Also this method is very popular as one of the bulk mining methods to mine large low-grade ore bodies.

Main elements of the system/working parameters:

- Diving deposit in levels with level interval in the range of 50–90m all along strike extension
- Size (width) of main level and stope entries= 3–7m.
- Height of opening= 2.7–4m (Based on equipment height).
- Length of stope= 50–90m (longitudinal); 25–40m (transverse).
- Minimum width of pillar in between stopes= 10m.
- Height of stope= level interval, range: 50–90m.

Unit operations

- Drilling: Development – Use of jacklegs, stopers and jumbo drills for drivage work. Stoping – drifter mounted ring and fan drilling rigs. DTH or ITH drills for large blast hole drilling.
- Blasting: use of NG based explosives, ANFO, slurries at the development headings. ANFO and slurries during stoping. In VCR method use of spherical charge consisting of high-density explosives. Charging with the use of pneumatic loaders. Firing – electrically or by detonating fuse. Secondary breaking at the stopes: plaster or pop shooting.
- Mucking: gravity flow of ore from stope to the extraction level horizon; mucking at the draw points with the application of LHD, FEL, rocker shovels.
- Haulage: LHD, trucks or tracked haulage system using mine cars and locomotive.

Advantages

- Productivity: Moderate to high, not labour intensive. Very good scope for mechanization
- Range of OMS: 15–30t/shift/man.
- Cost: Moderate (40% relative costs), low fragmentation and handling costs.
- Production rate: Moderate to high.
- Recovery: during stoping 85–95%, during pillar extraction – 60–80% Overall – upto 75%.
- Dilution: maximum up to 20%.
- Flexibility: unit operations can be carried simultaneously.
- Safety: little exposure to unsafe conditions, workers work under the protective roofs, easy to ventilate, thus better working conditions. Limitations
- Development cost and time: development cost fairly high. Stope preparation takes considerable time. Development and rock fragmentation costs during stoping increases when the deposit thickness is small and ore strength is high.
- Damages by heavy blasting: noise, vibrations, air blast and structural damages.
- Skill: requires special skill for accurate drilling otherwise hole deviation is common.
- Capital required: sufficient

ii. Supported Mining Methods

Supported mining methods are often used in mines with weak rock structure.

a) Cut and fill mining method

✓It is one of the more popular methods used for vein deposits and has recently grown in use.

✓It is an expensive but selective mining method, with low ore loss and dilution. (i.e., allows selective mining and avoid mining of waste or low grade ore).

✓It is a method of **short hole mining used in steeply dipping or irregular ore zones and scattered mineralization, in particular where the hangingwall limits the use of long hole methods.**

✓It requires **working at face.**

It is the most common of supported mining methods and is used:-

- in mining steeply dipping orebodies in stable rock masses (primarily in steeply dipping metal deposits), strata with good to moderate stability, and comparatively high grade mineralization.
- Either fill option may be consolidated with concrete, or left unconsolidated.
- Bottom up mining method: Remove ore in horizontal slices, *starting from a bottom undercut and advancing upward .*
- Moderate production rates.
- Good resource usage.
- Ore is drilled, blasted and removed from stope.
- The ore is mined in slices: As each horizontal or slightly inclined slice is taken, the **voids (Opens) are *backfilled with a variety of fill types to support the walls. (i.e., the fill can be rock waste, tailings, cemented tailings, or other suitable materials)***

Ore is drilled, blasted and removed from stope

Stopes

"Fill" is some combination of tailings and cement

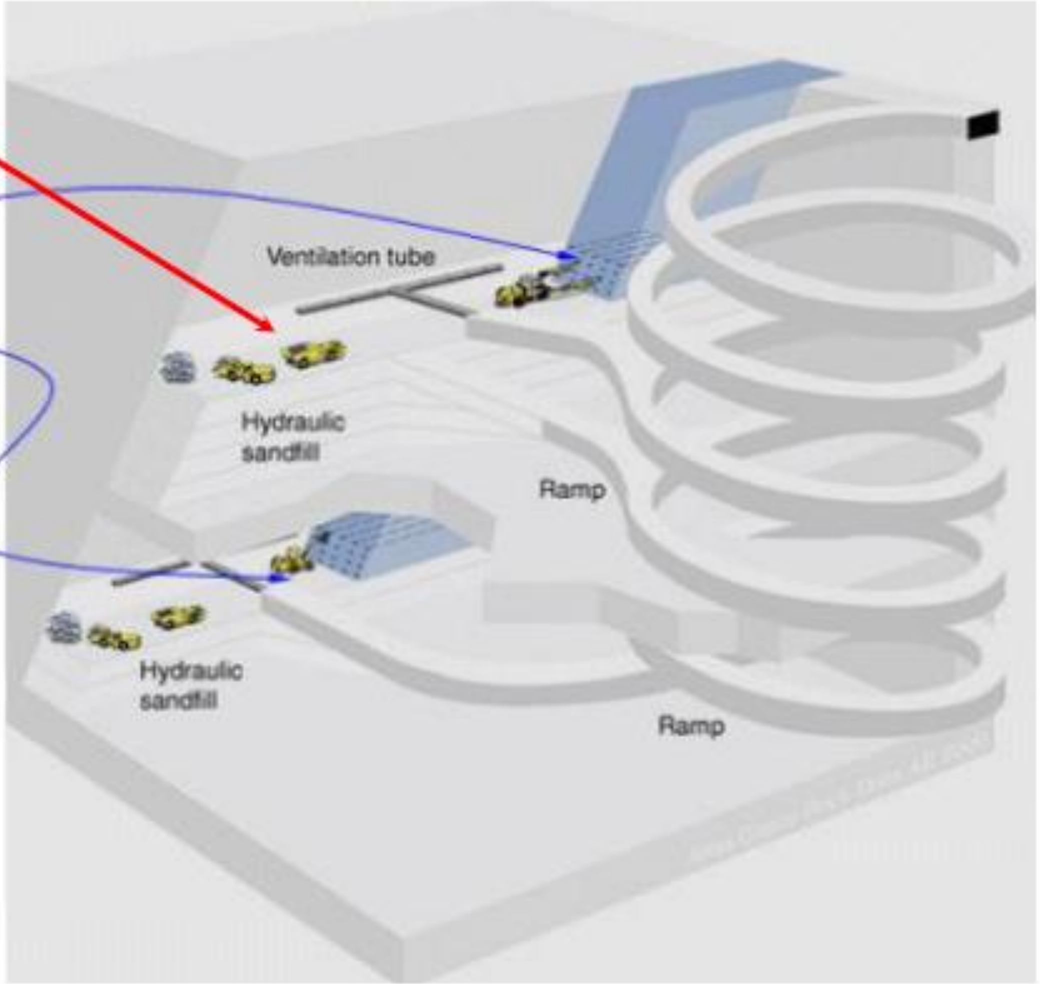


Figure showing cut and fill mining method.

Remove ore in horizontal slices, *starting from a bottom undercut and advancing upward*.

As each horizontal slice is taken, the **voids (Opens)** are *filled with a variety of fill types to support the walls.* (i.e., the fill can be rock waste, tailings, cemented tailings, or other suitable materials).

- Because the method involves moving fill material as well as a significant amount of drilling and blasting, *it is relatively expensive and therefore is done only in high grade mineralization* where there is a need to be selective and avoid mining of waste or low grade ore.
- It is practiced both in the ***overhand (upward) and in the underhand (downward) directions***.

i) Overhand (upward) cut and fill

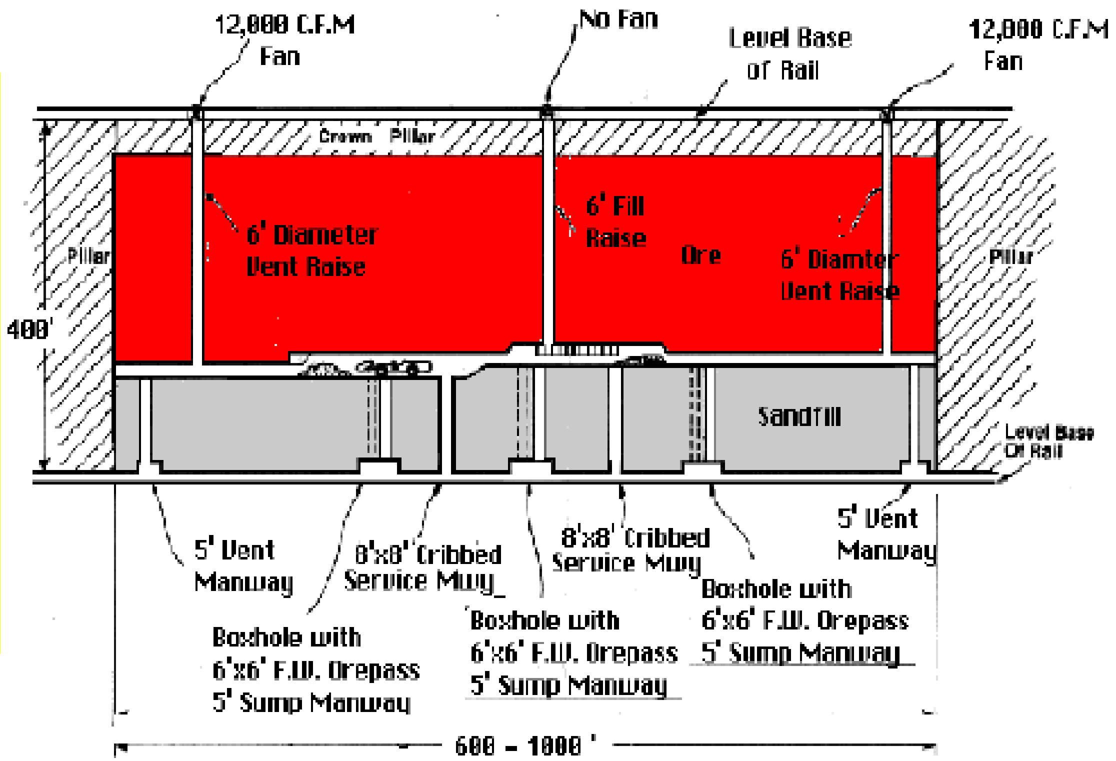
- is applied to ore lies underneath the working area and the roof is backfill.
- involves a work area of cemented backfill while mining ore from the roof.

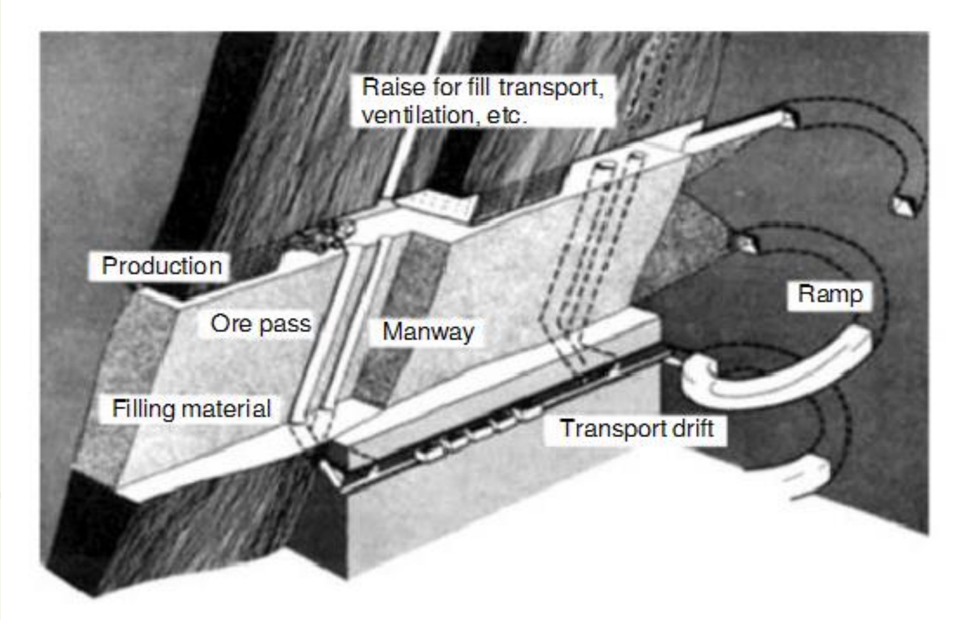
ii) Underhand (Downward) cut and fill ore

- is applied to ore lies beneath the working area and the roof is cemented backfill.
- ore overlies the working area and the machines work on backfill.

Note:

Drift and Fill is similar to cut and fill, except it is used in ore zones which are wider than the method of drifting will allow to be mined. In this case the first drift is developed in the ore, and is backfilled using consolidated fill. The second drift is driven adjacent to the first drift. This carries on until the ore zone is mined out to its full width, at which time the second cut is started atop of the first cut.





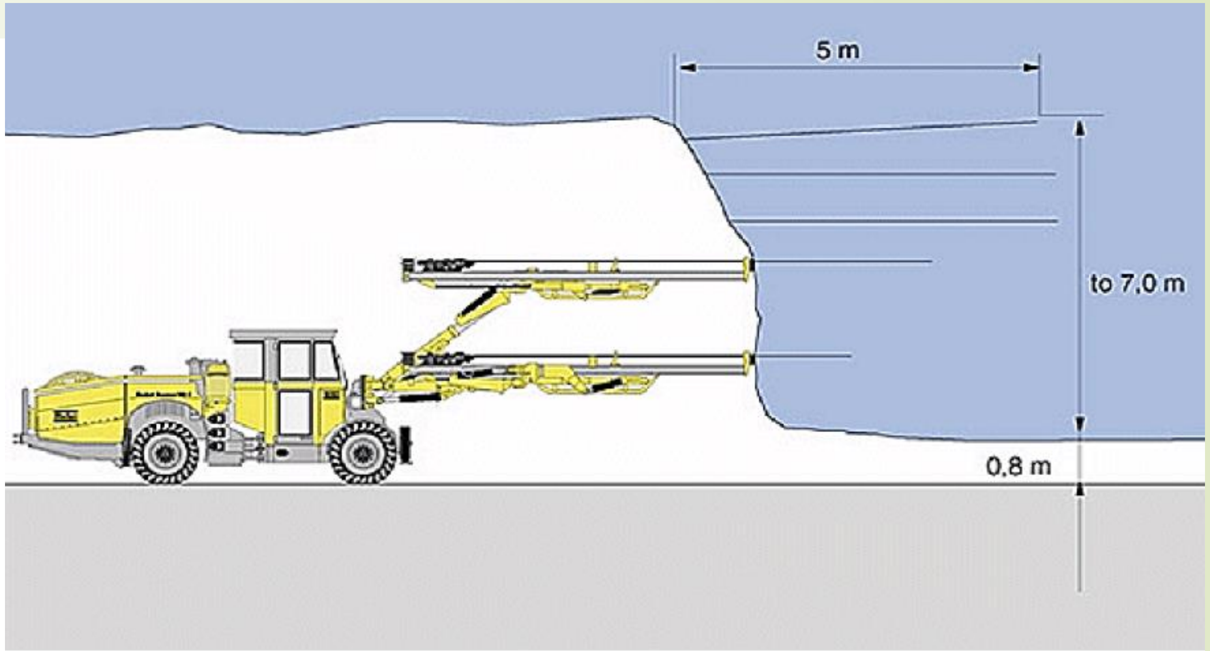
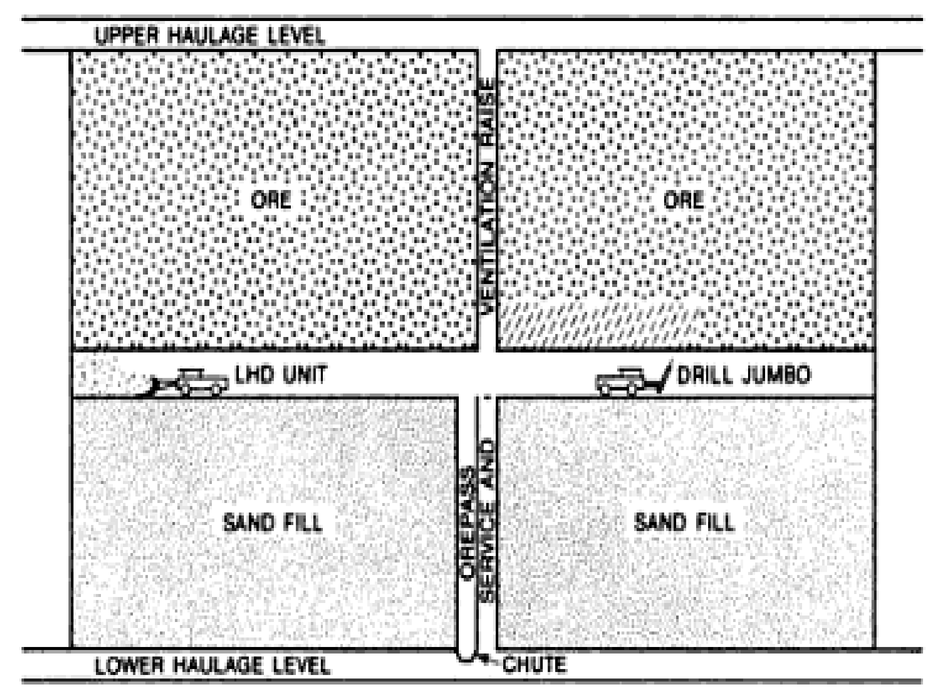


Figure shows "Slim-size" machines including drill rigs, jumbos, and 2 m³ bucket **LHDs**, are available for working in drifts as narrow as 2 m.



Advantages

- Productivity: moderate 10–20 tons./man-shift, maximum up to 30–40t/man/shift.
- Production rate – moderate. But can be applicable for small mines and even for irregular ore bodies.
- Scope of mechanization – moderate.
- Safety – good safety records. Proved safer even at great depths and prevents occurrence of rock bursts.
- Flexibility: versatile, adaptive to variety of conditions. If grade of ore is poor selective mining and ore sorting is possible.
- Ground conditions – suitable even for the worst ground conditions.
- Capital required – moderate.
- Stope development – little.
- Recovery – maximum if pillar mined, it could be up to 95% or more, and ore fines losses are eliminated. Low dilution.
- Stope development – little.
- Depth – proved vital for deep mining at high rock pressures.

Following are the suitable conditions, in general, for the application of cut & fill stoping:

- Ore strength: moderate to strong.
- Rock strength: weak.
- Deposits shape: any, regular to irregular.
- Deposit's dip: usually steep but can be applied for flat dips also then it will be similar to longwall mining.
- Size and thickness: fairly large extent, thin to thick (2–30m).
- Ore grade: high but uniformity can be variable.
- Depth: practiced up to 2.5km

Limitations

- Mining cost – relative cost 60%.
- Cost of backfilling – up to 50% of the total mining cost.
- Operational skill – requires skilled labour. It is more of labour intensive.
- Working atmosphere: at depth wet filling may create humidity problems.

Variants

- Cut and Fill with flat back – (i) Conventional; (ii) Mechanized
- Cut and Fill with inclined slicing
- Longwall Cut and Fill stoping
- Post & Pillar – Cut and Fill stoping
- Stope-drive cut and fill stoping: (i) starting from upper level (ii) starting from lower level. Selection of a particular variant of cut and fill stoping would depend upon, the strata conditions.

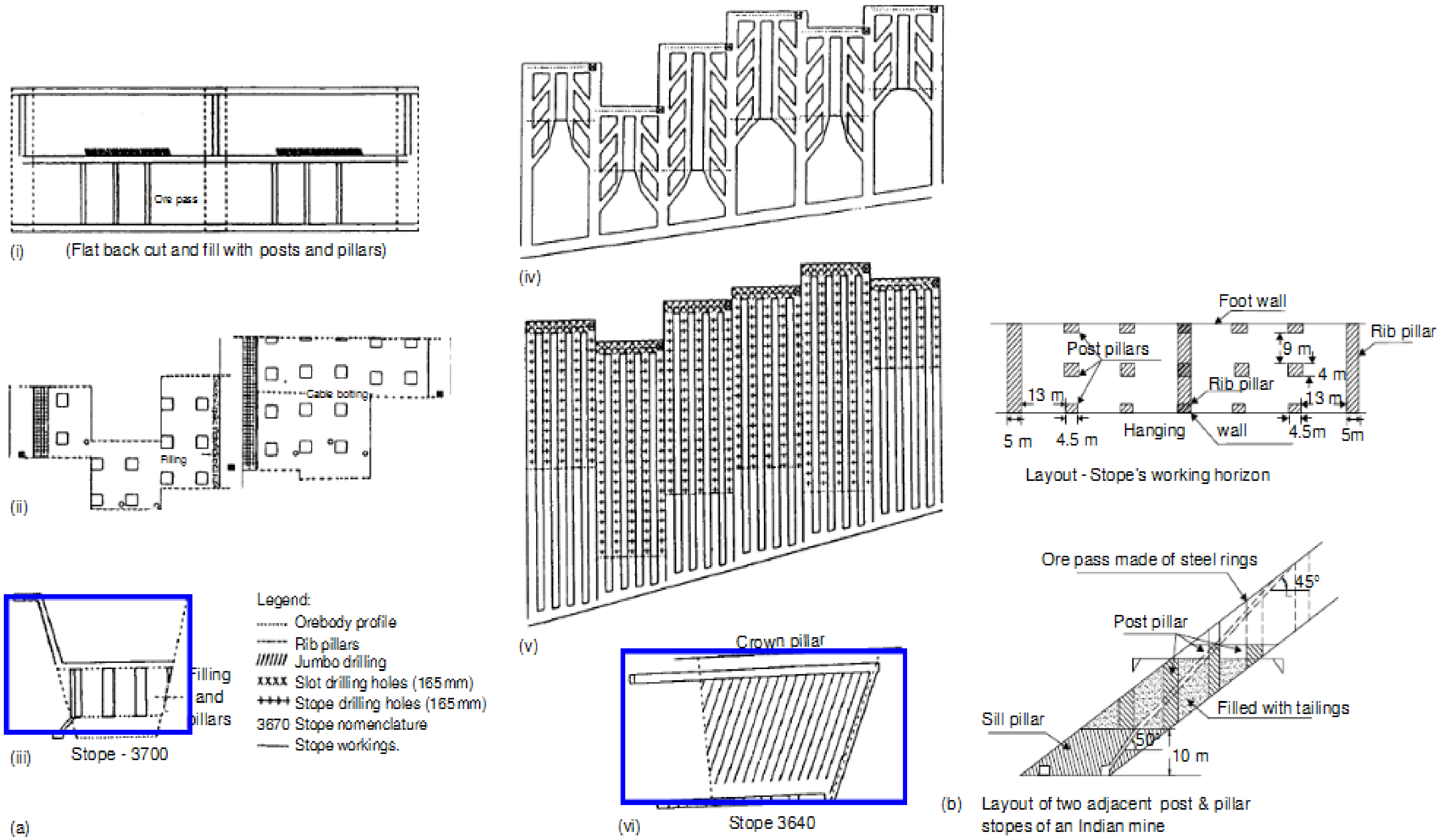


Figure: (a) showing layout details of a post and pillar cut and fill stope for mining weak and wider Pb-Zn deposits. (b) Post and pillar layout of a copper deposit

b) Square-set stoping

- The square-set stoping method is used where the ore is weak, and the walls are not strong enough to support themselves.
- The value of the ore must be relatively high, for square-setting is slow, expensive, and requires highly skilled miners and supervisors.
- In square-set stoping, one small block of ore is removed and replaced by a "set" or *cubic frame of timber which is immediately set into place.*
- The timber sets interlock and are filled with broken waste rock or sand fill, for they are not strong enough to support the stope walls.
- The waste rock or sand fill is usually added after one tier of sets, or stope cut, is made.
- *Square-set stoping* also involves backfilling mine voids; however, it relies mainly on timber sets to support the walls during mining.
- This mining method is rapidly disappearing in North America because of the high cost of labor.
- However, it still finds occasional use in mining high-grade ores or in countries where labor costs are low.

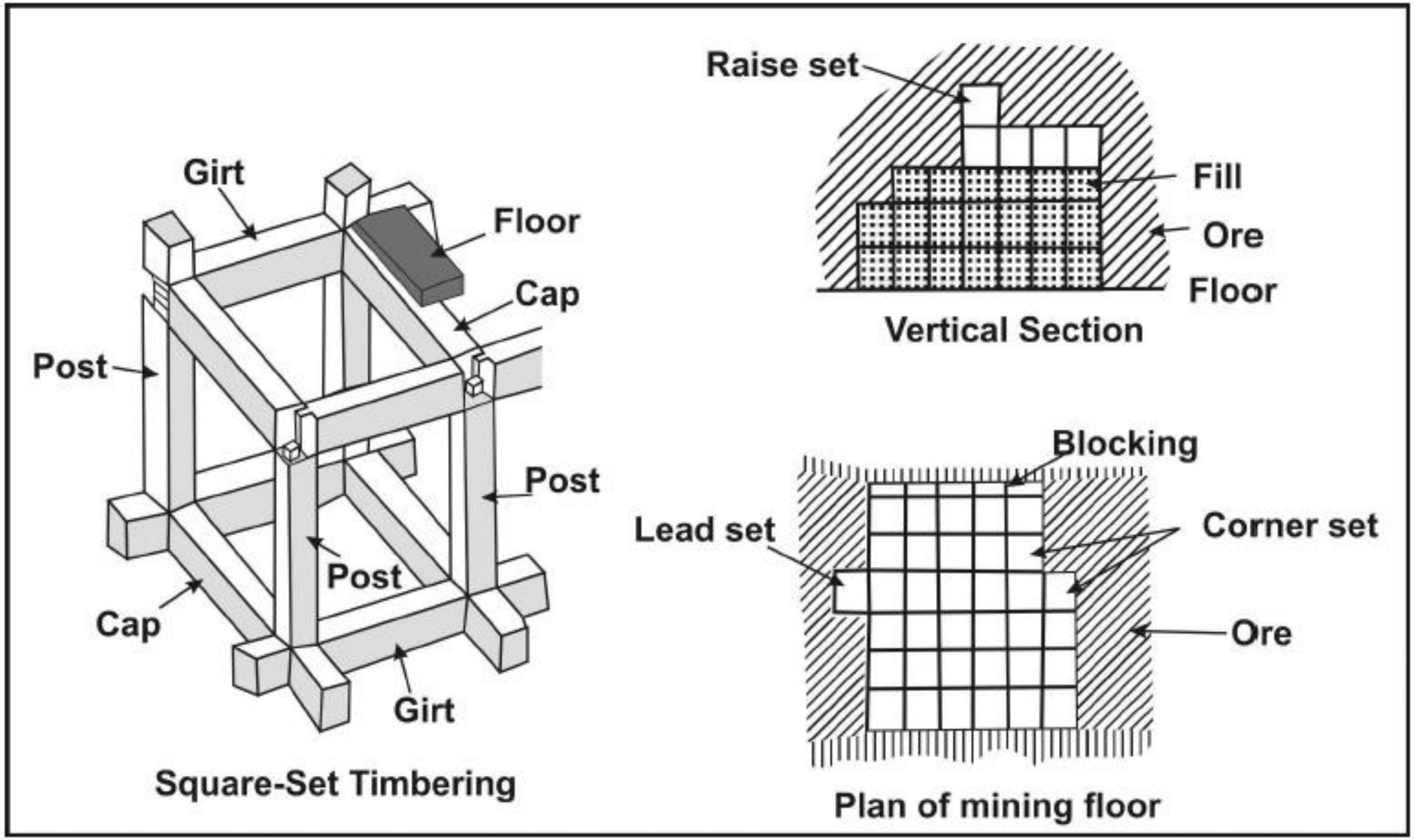


Figure shows Square-set stoping

c) Stull stoping

- is a supported mining method using timber or rock bolts in tabular, pitching ore bodies.
- is one of the methods that can be applied to ore bodies that have dips between 10º and 45º.
- often utilizes artificial pillars of waste to support the roof.

Supported Mining Methods		
Factor	**Cut and Fill Mining**	**Square Set Stoping**
Ore strength	Moderate \ strong	weak
Rock strength	Weak	weak
Deposit shape	Tap \ irregular	Any
Deposit dip	Fairly steep	Any
Deposit size	Thin \ Moderate	Usually small
Ore grade	Fairly high	High
Ore uniformity	Variable	Variable
Depth	Moderate \ deep	Deep

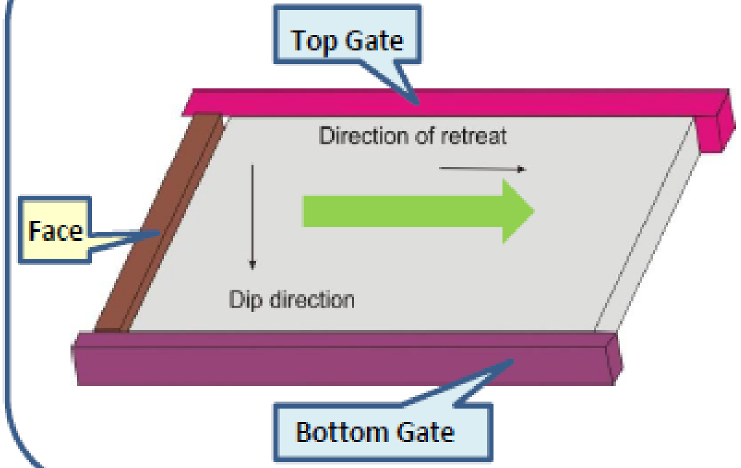
2. Caving (or Bulk) Mining methods:

- Caving methods are varied and versatile and involve caving the ore and/or the overlying rock.
- Subsidence of the surface normally occurs afterward

a) Longwall stoping

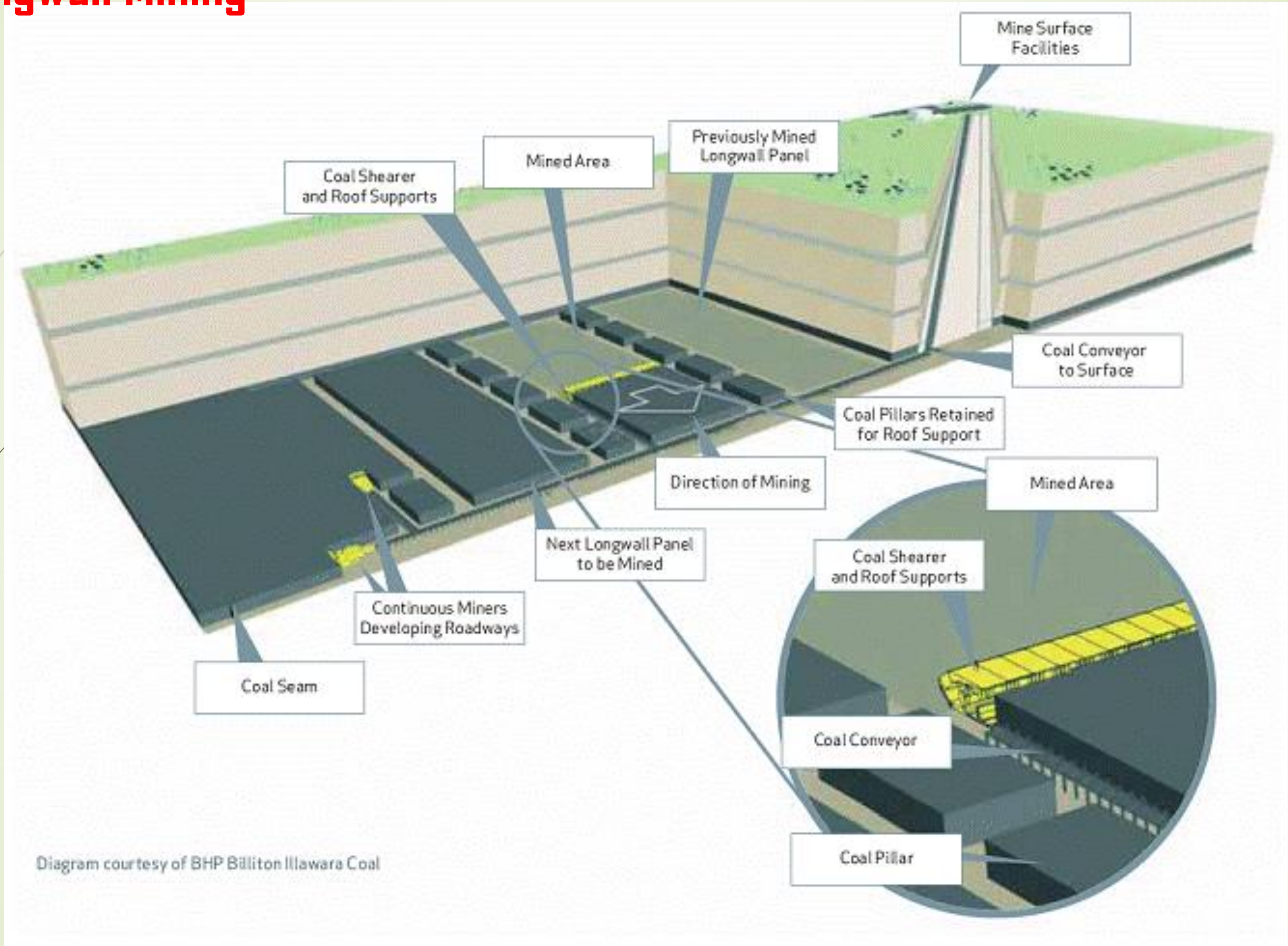
- is a caving method particularly well adapted to relatively flat-lying, thin, planar deposits or horizontal seams, usually ***coal***, at some depth.
- *is suitable for tabular orebodies, with moderate dip* (e.g., coal and stratiform hard-rock ores like diamond deposits).
- Need to divide orebody to "face" or the "working face“. The collection of *cuts, cross-cuts, and pillars* all together make up a "panel" and all the equipment that goes together to operate in that panel is a "unit or Longwall units".
- In this method, a face of considerable length (a long face or wall) is maintained, and as the mining progresses, the overlying strata are caved, thus promoting the breakage of the coal itself.
- Applied to longer (~100 m) and longer diameter blastholes (*i.e.*, *thus requiring less drilling than sublevel stoping*).
- Greater drilling accuracy is required.
- Need to a ***longwall machine*** (It's *designed* to let the roof fall behind it, and mines out big rooms in which the roof almost immediately collapses, leaving only a small entryway and the metal barrier that protects the longwall unit).
- Traditionally high production rates.
- Large openings with long open times.
- High ground support cost .
- *Bottom up mining method.*
- *Non-selective mining.*
- Not stress friendly.
- Many equipment types.

LONGWALL GENERAL LAYOUT



Longwall mining method includes drivage of two long roadways in coal and joining them at the end by a perpendicular drivage forming a face.

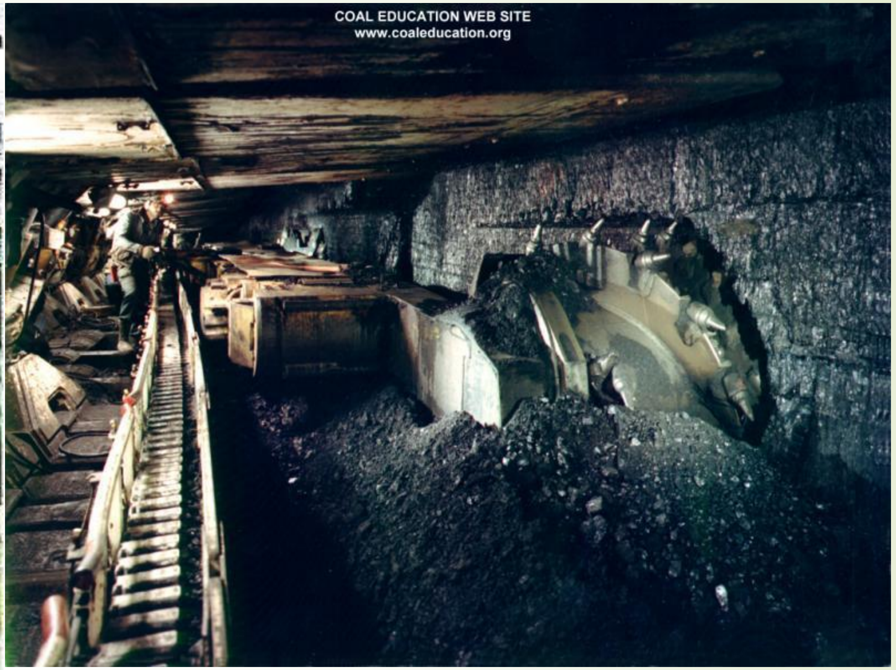
Longwall Mining



Example: Longwall Mining of Coal

- Longwall mining is a highly mechanized underground mining system for mining coal.
- It set of longwall mining equipment consists of a coal shearer mounted on conveyor operating underneath a series of self-advancing hydraulic roof supports.
- Almost the entire process can be automated.
- **Longwall mining machines are typically 150-250 meters in width and 1.5 to 3 meters high**.
- Longwall miners extract "***panels***" - rectangular blocks of coal as wide as the face the equipment is installed in, and as long as several kilometers.
- A layer of coal is selected and blocked out into an area known as a ***panel*** **(A typical panel might be 3000 m long X 250 m wide).**
- ***Passageways*** would be excavated along the length of the panel to provide access and to place a conveying system to transport material out of the mine.
- Entry tunnels would be constructed from the passageways along the width of the panel.
- **Extraction is an almost continuous operation involving the use of:** self-advancing hydraulic roof supports sometimes called ***shields***, ***a shearing machine***, and *a* ***conveyor which runs parallel to the face being mined***.
- Powerful mechanical coal cutters (***Shearers***) cut coal from the face, which falls onto an armoured face conveyor for removal.
- The longwall system would mine between entry tunnels.
- Longwalls can ***advance*** *into an area of coal, or more commonly, retreat back between development tunnels* (called "***Gate roads***")
- As a longwall miner ***retreats back along a panel***, *the roof behind the supports is allowed to collapse in a planned and controlled manner.*

Longwall Mining Machine



It's *designed to let the roof fall behind it, and mines out big rooms in which the roof almost immediately collapses, leaving only a small entryway and the metal barrier that protects the longwall unit.*

b) Sublevel Caving Stoping

- It is used to mine large orebodies with steep dip tabular or massive deposit and continuation at depth.

- The ore is extracted via sublevels which are developed in the orebody at regular vertical spacing.

- Each sublevel has a systematic layout of parallel drifts, along or across the orebody.

- **Sublevel stoping recovers the ore from open stopes separated by access drifts each connected to a ramp.**

- The orebody is divided into sections about 100 m high and further divided laterally into alternating stopes and pillars.

- A main haulage drive is created in the footwall at the bottom, with cut-outs for draw-points connected to the stopes above. The bottom is V-shaped to funnel the blasted material into the draw-points.

- ***Longhole rigs drill the ore section above a drift. The ore in the stope is blasted, collected in the draw-points, and hauled away.***

- Blasting on each sublevel *starts at the hangingwall and mining then proceeds toward the footwall.*
- **Blasting removes support for the hangingwall, *which collapses into the drift.***
- As mining progresses *downward, each new level is caved into the mine openings, with the ore materials being recovered while the rock remains behind.*
- Loading continues until it is decided that waste dilution is too high *Work then begins on a nearby drift heading with a fresh cave.*
- As mining removes rock ***without backfilling, the hangingwall keeps caving into the void. Continued mining results in subsidence of the surface, causing sink holes to appear. Ultimately, the ground surface on top of the orebody subsides (Fig.18).***
- However, the stopes are ***normally backfilled with consolidated mill tailings after being mined out (This allows for recovery of the pillars of unmined ore between the stopes, enabling a very high recovery of the orebody).***

Main elements of the system/working parameters– An optimum range of these parameters could be as under:

- Dividing deposit in levels spaced 60–80m apart, all along the strike extension.
- Size of main level and stope entries= 3–5m.
- Height of opening= 2.7–4m (Based on equipment height).
- Length of stope= 50–90m (longitudinal), 50–60m (transverse).
- Height of stope= level interval, range: 60–80m.Details of the sublevel entries:
- Distance between sublevels vertically= from 9–14m to 20–32m
- Spacing between crosscuts (if transverse stoping) from 7.5–11m to 23m (centre to centre)
- Dimensions of drill crosscuts (drives) – width 3m–6m height 2.5m–4m
- Fan inclination= vertical to 70–80°(figs 16.26 (e) and (f)), Inclination of outer holes 70–85°.
- Ring burden= 1.2–1.8m.

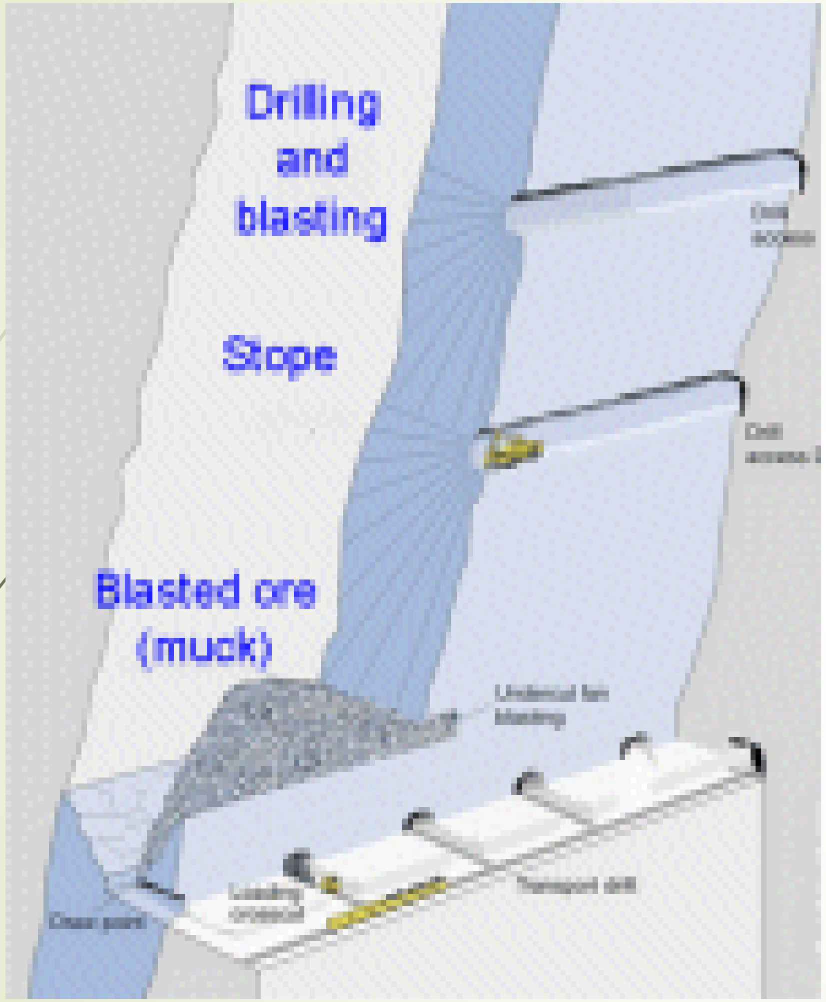


Figure : Sublevel stoping

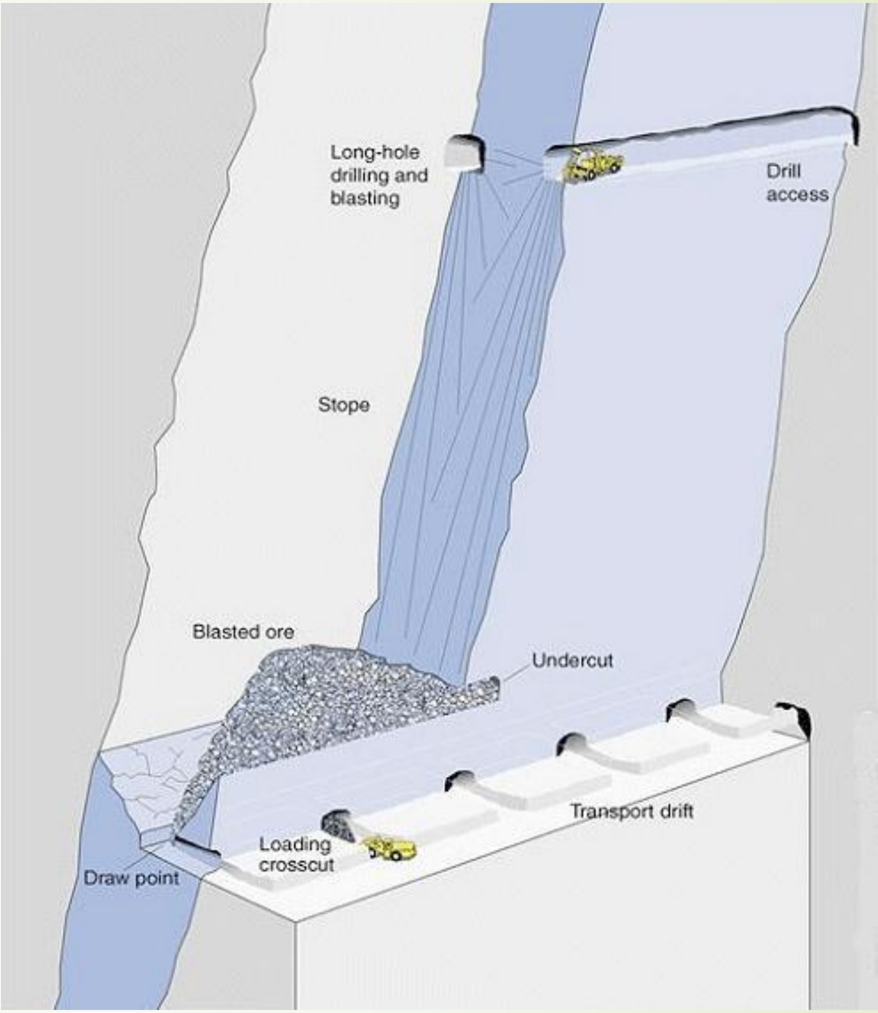


Fig. : longhole stoping and removing ore after blasting

Sublevel Caving

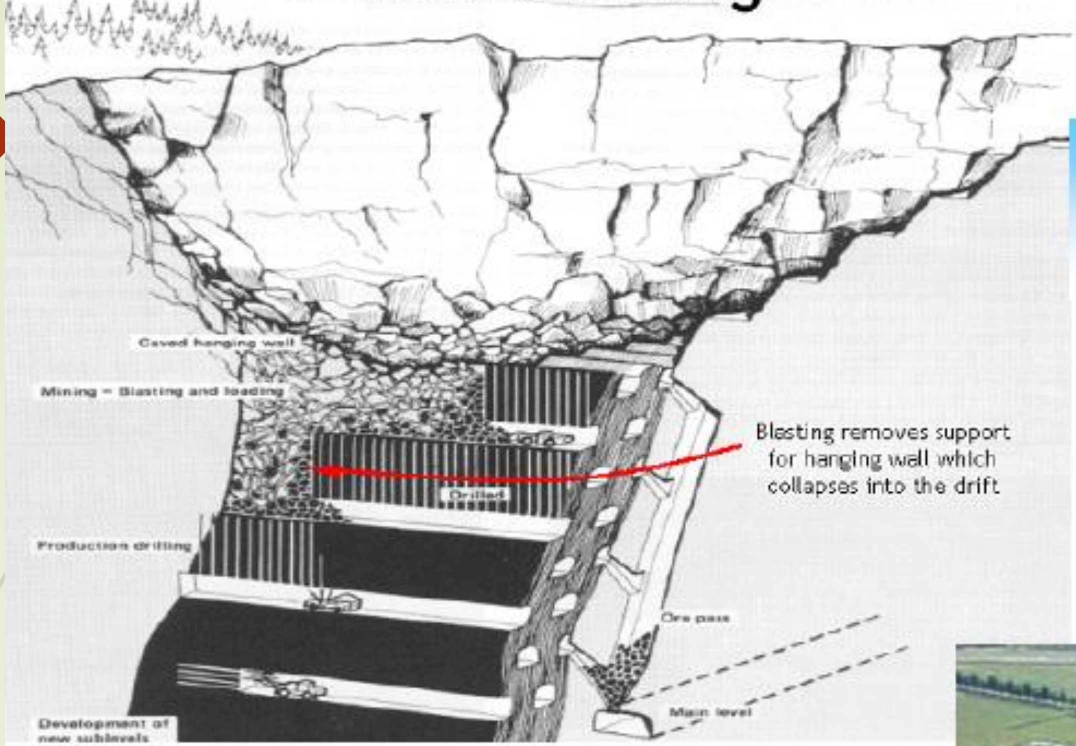


Figure shows sublevel caving is used to mine large orebodies with steep dip and continuation at depth.

Figure shows Continued mining results in subsidence of the surface, causing sink holes to appear.

Ultimately, the ground surface on top of the orebody subsides.

Advantages

- Productivity: Fairly high, OMS in the range of 20–40t/shift/man.
- Production rate – High. It can be considered as one of the bulk mining methods.
- Recovery – with pillar extraction 80–90%, but with dilution sometimes it exceeds 100%.
- Commencement of regular production; at an early stage even during stope development 20–25% of the designed production rate can be achieved. Stope development and stoping activities goes simultaneously.
- Adaptability to mechanization: suitable for varying degree of mechanization.
- Safety – little exposure to unsafe conditions, workers work under the protective roofs, easy to ventilate, thus better working conditions.

Limitations

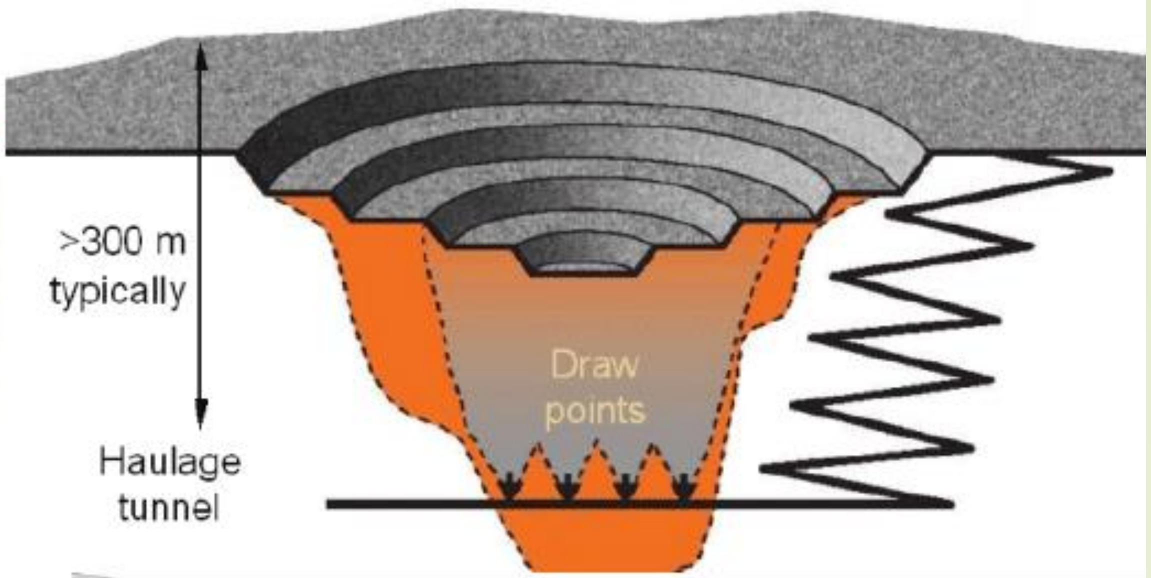
- Dilution is high, ranging from 10 to 35%, and that results considerable ore losses in the stopes. This feature also limits its application to high valued ores.
- Subsidence and caving occur over wide areas causing land degradation.
- Cost is comparatively high (40–60% relative cost). Development cost and time are fairly high. Stope development is responsible for the high costs. Capital required –sufficient.
- Provision for the equipment access in the stope needs to be incorporated

c) Block Caving stoping

- Block-caving method ***is employed generally for steeply dipping ores, and thick sub-horizontal seams of ore.*** The method has application, for example in sulphide deposits and underground kimberlite (diamond) mining.
- It is most applicable to :-
  - A large-scale or bulk mining method that is highly productive, low in cost, and used primarily on massive steeply dipping orebodies that must be mined underground.
  - Weak or moderately strong orebodies that readily break up when caved.
  - Large, deep (>two km deep), low-grade deposits with high friability (Fig.19
- It is often done to continue mining *after open pit mining becomes uneconomic or impossible*. However, some mines start as block cave operations (e.g., *There are several of these in Chile. Rio Tinto is considering a deep at the Resolution deposit to the east of Phoenix).*
- A grid of tunnels is driven under the orebody → ***The rock mass is then undercut by blasting***.
- Ideally the rock will break under its own weight → Broken ore is then taken from draw points.
- There may be hundreds of draw points in a large block cave operation (Fig.20).

- An *undercut with haulage access is driven under the orebody*, with "***drawbells***" excavated between the top of the haulage level and the bottom of the undercut. *The drawbells serve as a place for caving rock to fall into.*
- The orebody is drilled and blasted *above* the undercut, and the ore is removed via the haulage access.
- Due to the friability of the orebody the ore above the first blast caves and falls into the drawbells. As ore is removed from the drawbells *the orebody caves in providing a steady stream of ore*[3].
- If caving stops and removal of ore from the drawbells continues, *a large void may form, resulting in the potential for a sudden and massive collapse and potentially catastrophic windblast throughout the mine.*[4]
- Where caving does continue, the ground surface may collapse into a surface depression such as those at the Climax and Henderson molybdenum mines in Colorado. Such a configuration is one of several to which miners apply the term "***glory hole***".
- Orebodies that do not cave readily are sometimes preconditioned by *hydraulic fracturing, blasting, or by a combination of both*. Hydraulic fracturing has been applied to preconditioning strong roof rock over coal longwall panels, and to inducing caving in both coal and hard rock mines.
- Essentially block caving creates an underground *'inverted open pit'*. **Surface subsidence can be a problem….???.**

Figure shows application of the Block caving to large, deep, low-grade deposits



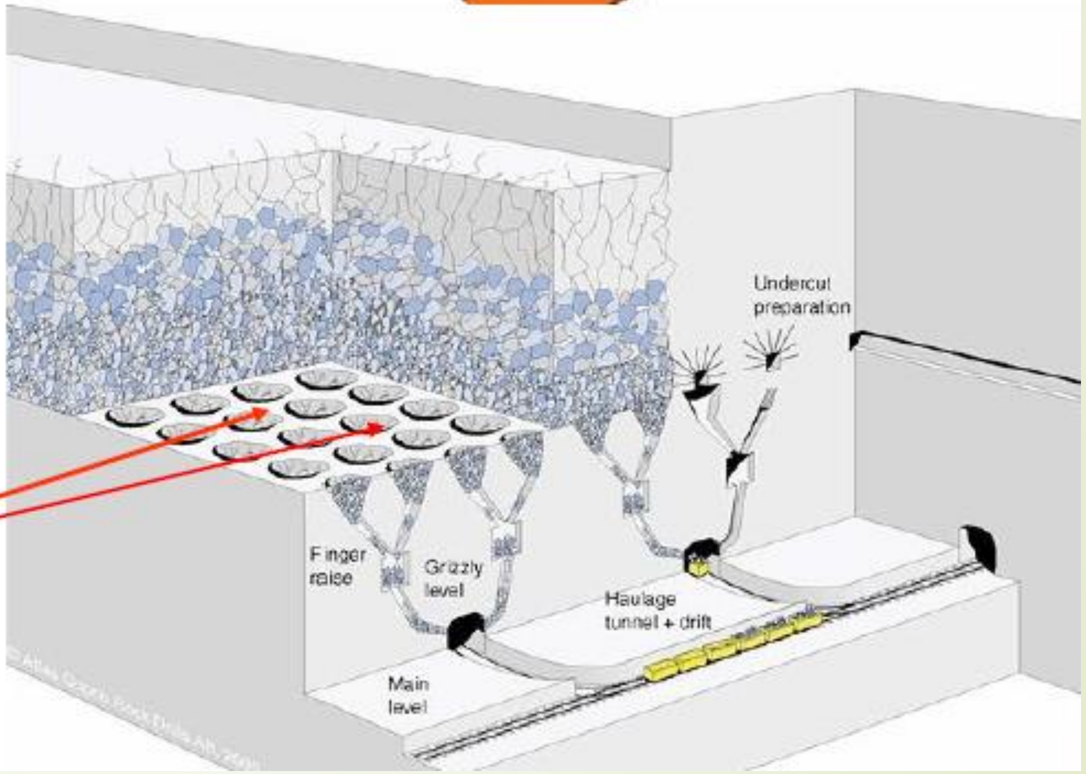
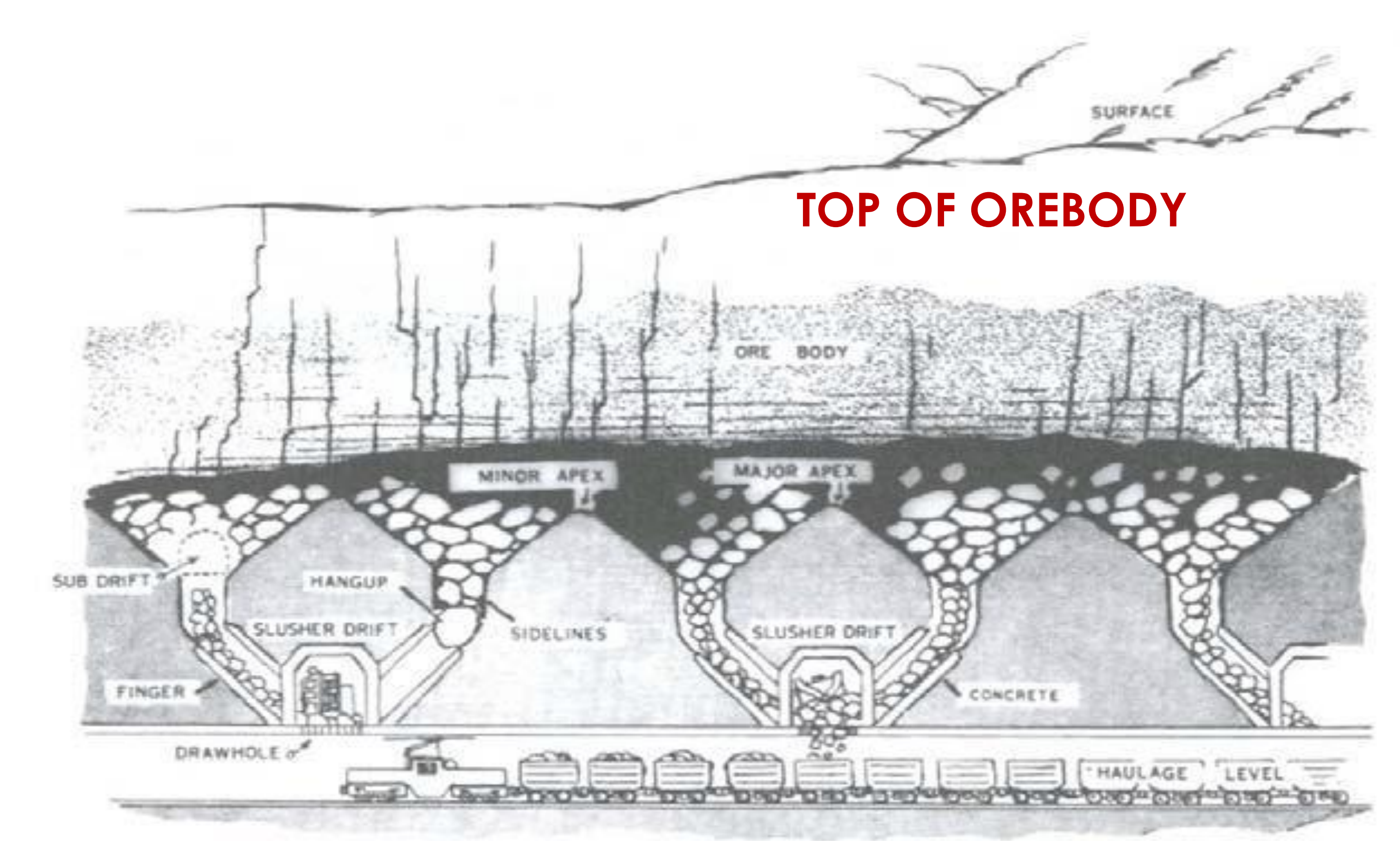


Figure shows hundreds of draw points to take broken ore in a large block cave operation



An undercut tunnel is driven under the orebody, with ***"drawbells" excavated above. Caving rock falls into the drawbells. The orebody is drilled and blasted above the undercut to initiate the "caving" process. As ore is continuously removed from the drawbells, the orebody continues to cave spontaneously, providing a steady stream of ore. If spontaneous caving stops, and removal of ore from the drawbells continues, a large void may form, resulting in the potential for a sudden and massive collapse and a potentially catastrophic windblast throughout the mine***

Note: *Both block caving and longwall mining are widely used because of their high productivity.*

Caving (or Bulk) Mining Methods		
Factor	**Longwall stoping**	**Block caving stoping**
Ore strength	Any	Weak \ Moderate
Rock strength	Weak \ Moderate	Weak \ Moderate
Deposit shape	Tabular	Massive \ thick
Deposit dip	Low \ flat	Fairly steep
Deposit size	Thin \wide	Very thick
Ore grade	Moderate	Low
Ore uniformity	Uniform	Uniform
Depth	Moderate \ deep	Moderate

Ore Removal

- In mines which use rubber tired equipment for coarse ore removal, the ore (or "muck") is removed from the stope (referred to as "mucked out" or "bogged") using center articulated vehicles (referred to as boggers or LHD [i.e., Load, Haul, Dump]). These pieces of equipment may operate using diesel or electric engines and resemble a low-profile front end loader.
- *In shallower mines* , the ore is then *dumped into a truck to be hauled to the surface*.
- In deeper mines the ore is *dumped down an ore pass* (a vertical or near vertical excavation) where it falls to a collection level. On the collection level, it may receive primary crushing via jaw or cone crusher. The ore is then moved by conveyor belts, trucks or occasionally trains to the shaft to be hoisted to the surface in buckets or skips and emptied into bins beneath the surface headframe for transport to the mill.
  - In some cases the underground primary crusher feeds an inclined conveyor belt which delivers ore via an incline shaft direct to the surface. The ore is fed down ore passes, with mining equipment accessing the ore body via a decline from surface.

UNIT OPERATIONS OF MINING

Ore extraction and underground development is achieved by precise drilling and blasting techniques.

- Drilling a pattern of holes into the rock.
- Charging (filling) the holes with explosive.
- Blasting the rocks.
- Bogging (digging) it out.
- Ground support.
- Transporting it to the surface.

During the development and exploitation stages of mining when natural materials are extracted from the earth, remarkably similar unit operations are normally employed.

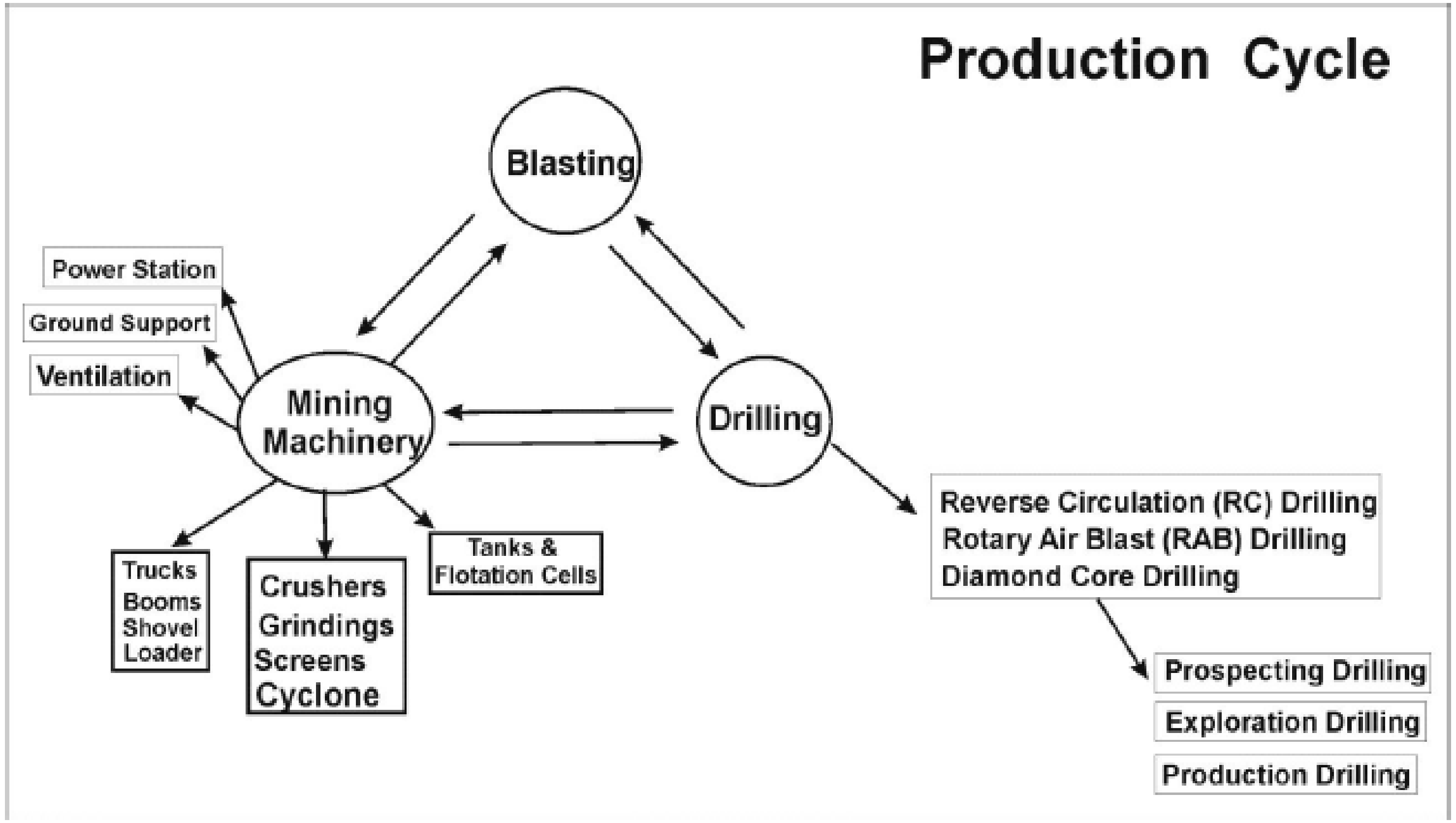
The *unit operations of mining* are the basic steps used to produce mineral from the deposit, and the auxiliary operations that are used to support them. The steps contributing directly to mineral extraction are production operations, which constitute the production cycle of operations.

The production cycle employs unit operations that are normally grouped into rock breakage and materials handling. Breakage generally consists of drilling and blasting and materials handling encompasses loading or excavation and haulage (horizontal transport) and sometimes hoisting (vertical or inclined transport).

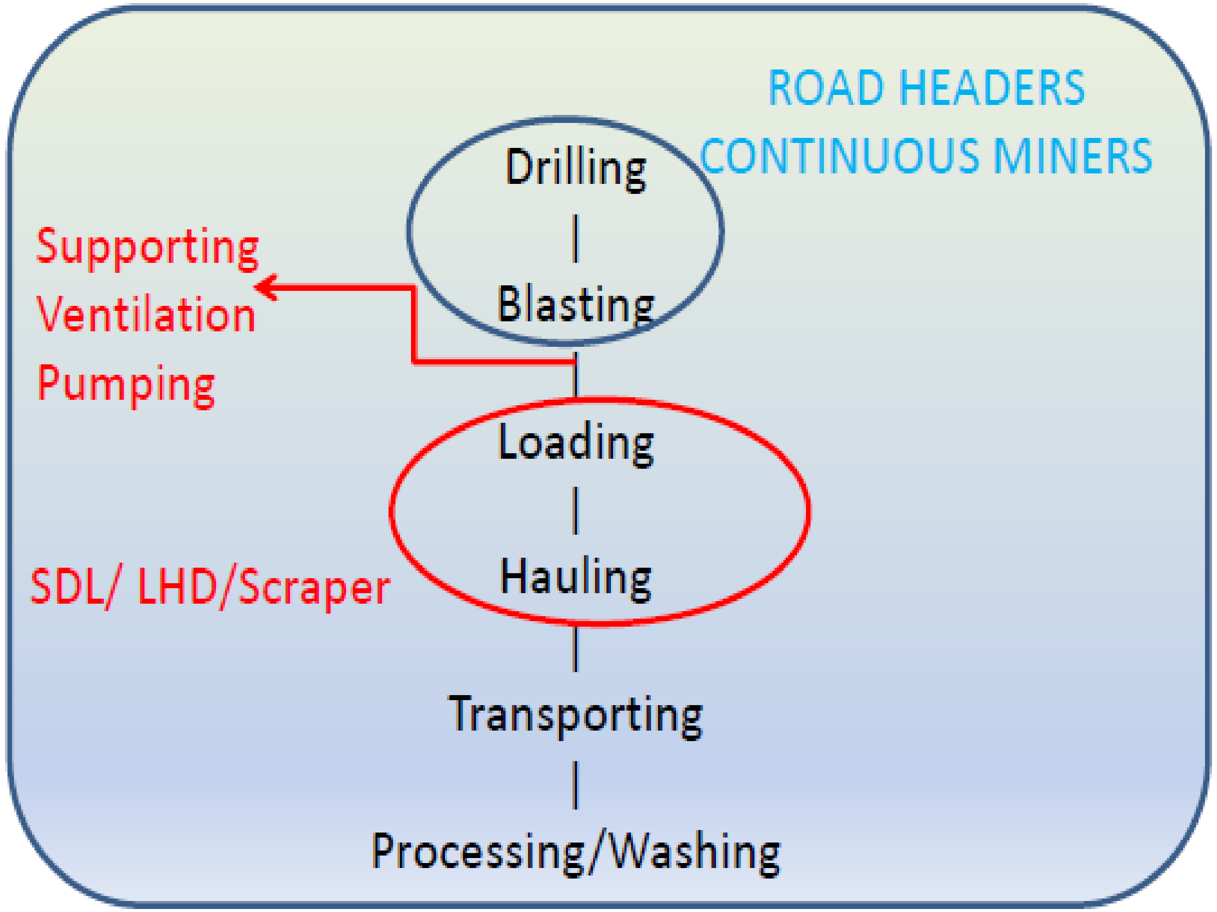
The Mining Production Cycle includes the following steps:-

- Drill holes.
- Blast.
- Mining machinery.
- Load .
- Haulage.
- Repeat until orebody is depleted.

Production Cycle =
Drill + Blast + Load + Haulage



Mining Process used in Underground Mining methods



Coal drill

|

Permitted explosives

|

Manual/ SDL/ LHD/Scraper

|

Manual/ SDL/ LHD/Scraper

|

Tubs, Conveyors, etc

|

Coal washeries

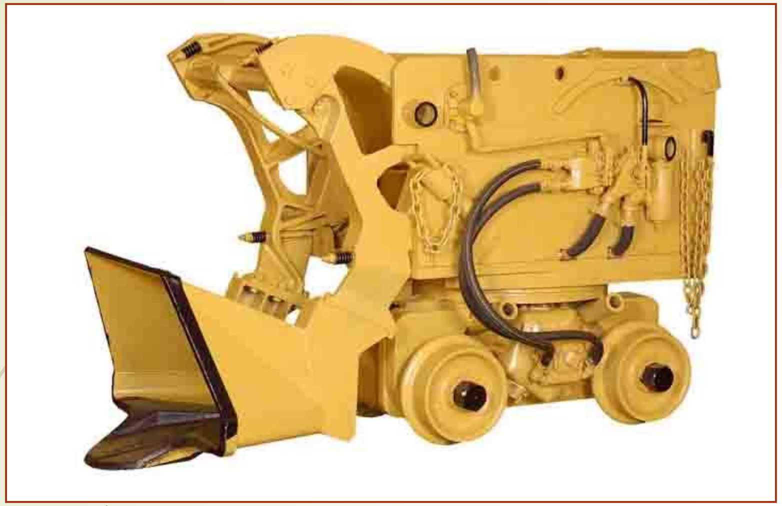
MMR 1961
CHAPTER-XI : Mine Workings

- 107 Underground workings (openings, ventilation, noxious gases, etc)
- 109. Workings under railways and roads, etc (45m in radious)
- 111. Working near mine boundaries (7.5m)
- 112. Support of workings (SSR)

General precautions – (1) Where several persons are working together in any place, one of them shall

- be placed in charge. No person shall be so appointed unless he is 21 years of age and has had not less than
- three years' experience in the workings of a mine.
- (2) No person shall work in any place other than his authorised working place.
- (3) Every person shall carefully examine his working place before commencing work and also at intervals
- during the shift. If any dangerous conditions is observed, he shall cease all work at that place and shall
- either take immediate steps to remove such danger or inform an official or the competent person in charge
- of the mine or district. Where several persons are working together and one of them is in charge, the
- examination required by this sub-regulation shall be made by the person in charge.
- (4) No person shall work or travel on any ledge or footpath less than 1.5 metres wide, from which he will
- be likely to fall more than 1.8 metres, unless he is protected by guard rails, fence or rope suitably fixed and
- sufficiently strong to prevent him from falling.
- (5) (a) No person shall carry or be permitted to carry any load along a road or footpath having an
- inclination of 30 degrees or more from the horizontal

Equipments



Loading equipments

Transportation equipments

Conclusive Remarks

- *A large number of working methods are practiced in underground hard rocks mines depending on the site specific conditions.*
- *However, for any given deposit the choice of a particular mining method depends on a number of factors.*
- *During the mine planning stage the major problem becomes how to decide the stoping method that will be most economic for a given deposit.*